

BIOLOGY

COMBINED SCIENCE NOTES BK 1 & 2



BIOLOGY SECTION

- **LABORATORY RULES AND SAFETY**
- **CELLS AND LEVELS OF ORGANISATION**
- **NUTRITION**
- **RESPIRATORY SYSTEMS**
- **TRANSPORT SYSTEM**
- **REPRODUCTION HEALTH AND DISEASE**

LABORATORY RULES AND SAFETY

LABORATORY SAFETY AND APPARATUS

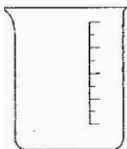
Laboratory rules

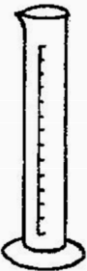
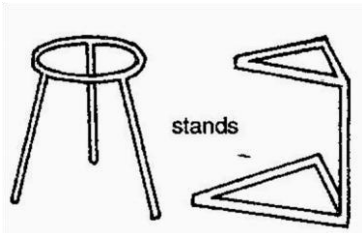
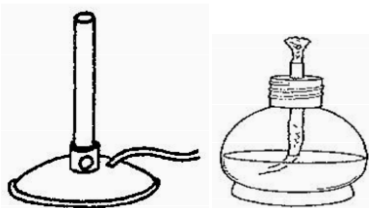
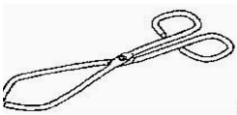
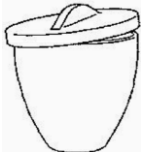
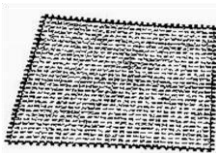
- A lab is a room equipped for scientific research, experimentation and analysis.
- Laboratory rules are a set of instructions that guide us in the laboratory.
- Safety in the laboratory is essential to prevent serious accidents to yourself and others, to avoid damage of apparatus, ensures accurate record keeping and to avoid injuries
- The following are a set of lab rules;
 - ✓ Do not work in the lab without your teacher
 - ✓ Read and follow instructions
 - ✓ Read labels on all chemical bottles before use
 - ✓ Do not use any chemical until the teacher has explained its uses and safety precaution to be taken
 - ✓ Do not perform any experiments and do not touch materials or apparatus without instructions from your teacher
 - ✓ Do not taste anything unless instructed to do so since some chemicals are toxic
 - ✓ Report all accidents to the teacher even if they appear to minor

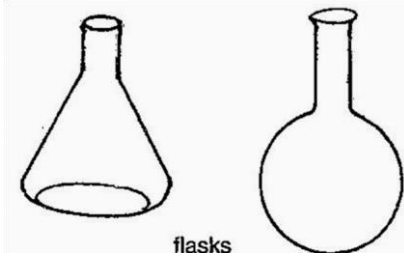
- ✓ In case of skin contact with chemicals, wash immediately with running water
- ✓ use burners with great care to avoid fires
- ✓ wash hands with soap after handling chemicals
- ✓ Clean your work areas and equipments after use
- ✓ Always wear safety clothing (lab coats, gloves & goggles)

Laboratory apparatus

- Apparatus refers to the set of equipments used by scientists to carry out a scientific investigation

Apparatus	Diagram	Function
Beaker		- used for heating, holding and measuring liquids though it is not accurate.
Test tube		- Hold small amounts of chemicals or liquids
Funnel		- Channels liquid or powdery solids from one container into another or can be used with filter paper to separate substances.

Measuring cylinder		- Is used to measure out the volume of liquids.
Tripod stand		- Have 3 legs and is used to support various apparatus above the flame during heating.
Burners		- Provides a flame used for heating substances
Tongs		- used to hold hot apparatus
Spatula		- Is used to scoop small amounts of solid chemicals from containers.
Crucible		- It is ceramic containers capable of withstanding extreme temperatures. They hold solids and small volumes of liquids for heating.
Wire gauze		- Is placed under a container that is being heated so that the container does not have direct contact with the flame. It spreads the heat from the burner and prevents glass from breaking

Evaporating dish		- used for evaporating solutions with or without heating
Flasks	 flasks	- used for heating, holding and measuring liquids though it is not very accurate

CELLS AND LEVELS OF ORGANISATION

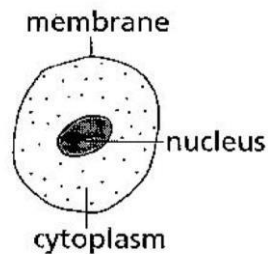
CELLS PLANT AND ANIMAL CELLS

- A cell is the basic unit building block of all living things.
- It is the fundamental unit of structure and function of life. It may perform a specific function in the body of a living organism.

- Living things can be classified as multi-cellular i.e. are made of many cells or unicellular organisms that are composed of only a single cell.

Structure of a plant and animal

Animal cell

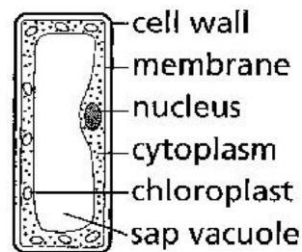


Cell membrane
around the

Cytoplasm

Nucleus

Plant cell



It is a partially permeable layer that forms a boundary

cytoplasm. It retains the cell contents and controls the movement of substances into and out of the cell. In this way the cell membrane maintains the structure and chemical reactions of the cytoplasm.

This is a jelly-like substance which contains particles and organelles. It is the site in which all chemical reactions take place in the cell.

It is always embedded in the cytoplasm. Control the type and quantity of enzymes produced by the cytoplasm. In this way it controls all the activities of the cell. It carries genetic information (DNA in form of chromosomes).

Cell wall

It is a tough non-living layer made of cellulose and other compounds surrounding the cell membranes. It confers shape and, to some extent, rigidity on the cells. It is permeable to water and other substances into and out of the cell. It prevents plant cells from bursting

Vacuole

It is a fluid filled space surrounded by a membrane. It contains cell sap containing salts and sugars dissolved in water. The outward pressure of the vacuole on cell wall makes the plant cells firm, giving strength and resilience to the tissues (keeps the cell rigid). Animal cells may sometimes have small vacuoles but are usually produced to do a particular job and are not permanent.

Chloroplasts

Contain the green pigment (chlorophyll) which absorbs light for photosynthesis.

Similarities and differences between plant and animal cells

Similarities

- All have cell membrane, cytoplasm and nucleus

Differences

- Plant cells are usually larger and their walls give them a distinct outline
- Plant cells are surrounded by a cell wall made of cellulose -
Often plant cells
have chloroplasts.
- Plant cells usually have a thin lining of cytoplasm, with a large central vacuole.

VARIATION

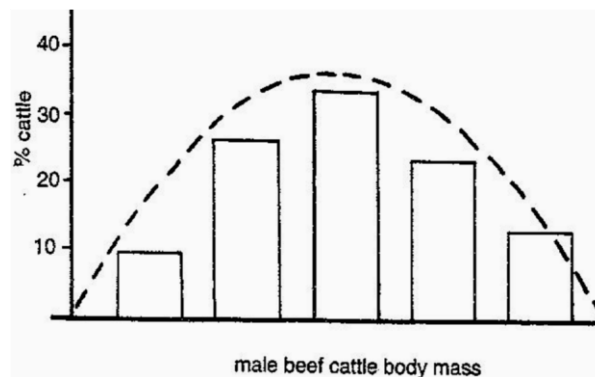
- variation is the observable difference among organisms with reference to specific characteristics.

- variations that can be inherited are determined by genes while other variations can be caused by the environment or a combination of both genes and the environment.
- Differences among organisms include height, sex, mass, earlobes, coat or skin colour, left or right handedness

Types of variation

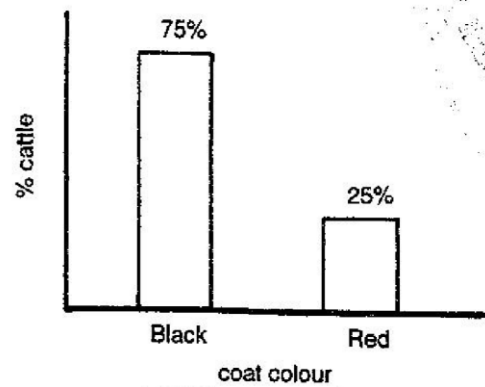
Continuous variation

- Show some intermediates and have no clear cut differences.
Occurs where differences between organisms show a gradual change. It is due to interactions between the environment and the genotype is most likely to be influenced by the environment e.g. height, mass, seeds in a pod, shoe size.



Discontinuous variation

- Occurs where there are clear cut differences organisms with no intermediates. Most features of discontinuous variation are clearly genetically determined and there is a limited number of alternatives e.g. tongue rolling, presence or absence of earlobes in humans, sex, eye colour, albinism, blood groups, coat colour.

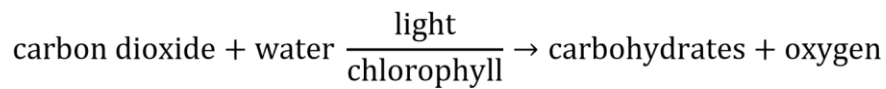


NUTRITION

PLANT NUTRITION

PHOTOSYNTHESIS

- Is the production of carbohydrates by green plants using carbon dioxide and water in the presence of light energy

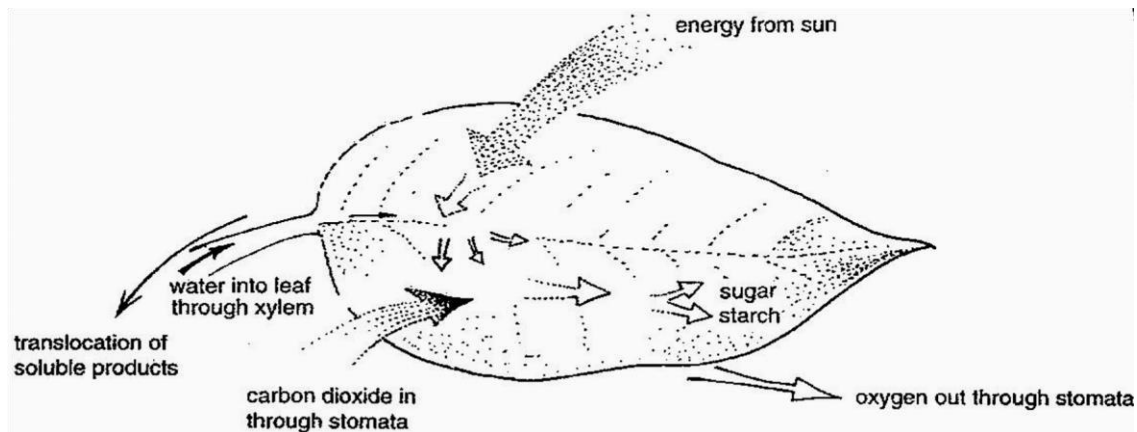


- Green plants convert light energy from the sun to chemical energy in carbohydrates during photosynthesis and later into other organic substances like starch.
- Plants also make loads of other nutrients for other organism to eat (consumers)

e.g. amino acids, vitamins, fats which are stored in leaves, roots, fruits and seeds.

- They produce oxygen needed by organisms for respiration.

- Green plants are autotrophy i.e. they make their own organic molecules from inorganic components. They are producers
- Animals are heterotrophy i.e. they feed on other organisms or their products to fuel life processes. They are consumers.



- Carbon dioxide and water enters the leaf cells. Chlorophyll in the chloroplasts traps light energy from the sun. the light energy is used to split water into hydrogen and oxygen (photolysis)

- The oxygen escapes from the leaf to the atmosphere as a by-product. The hydrogen reacts with carbon dioxide to form carbohydrates. Raw materials

1. Water is absorbed by root hairs and transported through the xylem

2. Carbon dioxide diffuses through the stomata Conditions needed

1. Chlorophyll essential to absorb solar energy
2. Sunlight End products

1. Carbohydrates can be used immediately by the plant or stored in the form of starch in seeds, fruits, stems and roots.
2. Oxygen diffuses out of the leaf or can be used for respiration by the plants and animals.

Importance of photosynthesis

- Produces food for all living organisms
- Produces oxygen which is used by plants and animals during respiration.
- Helps to reduce global warming by absorbing carbon dioxide
- Converts light energy into chemical energy

ANIMAL NUTRITION

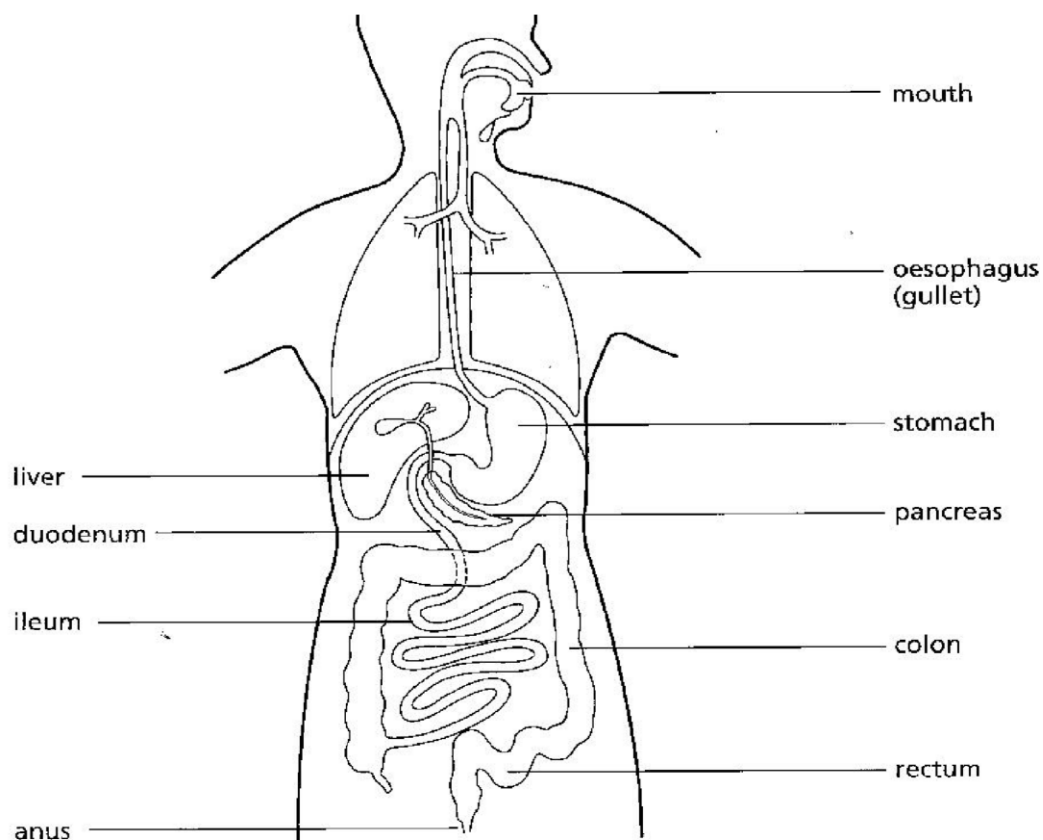
DIGESTIVE SYSTEM OF HUMANS

- The alimentary canal is a long tube which runs from the mouth to the anus. It is part of the digestive system which also includes liver, pancreas and gall bladder.

Parts of digestive system

- Food is taken into the digestive system through the mouth (ingestion). In the mouth the food is chemically digested by enzymes and mechanically digested by the teeth.
- When swallowed it passes through the esophagus and movement is facilitated by peristalsis.
- Food enters the stomach when it is churned for about 3-4 hours and mixed with digestive juices.

- Food enters the duodenum where there is chemical digestion of food by enzymes produced from the pancreas.
- In the ileum digestion is completed and nutrients absorbed into the blood stream by diffusion.
- Water and salts are absorbed in the colon by osmosis and active uptake.
- Waste matter is temporarily stored in the rectum and then passed out through the anus (Egestion).



BALANCED DIET

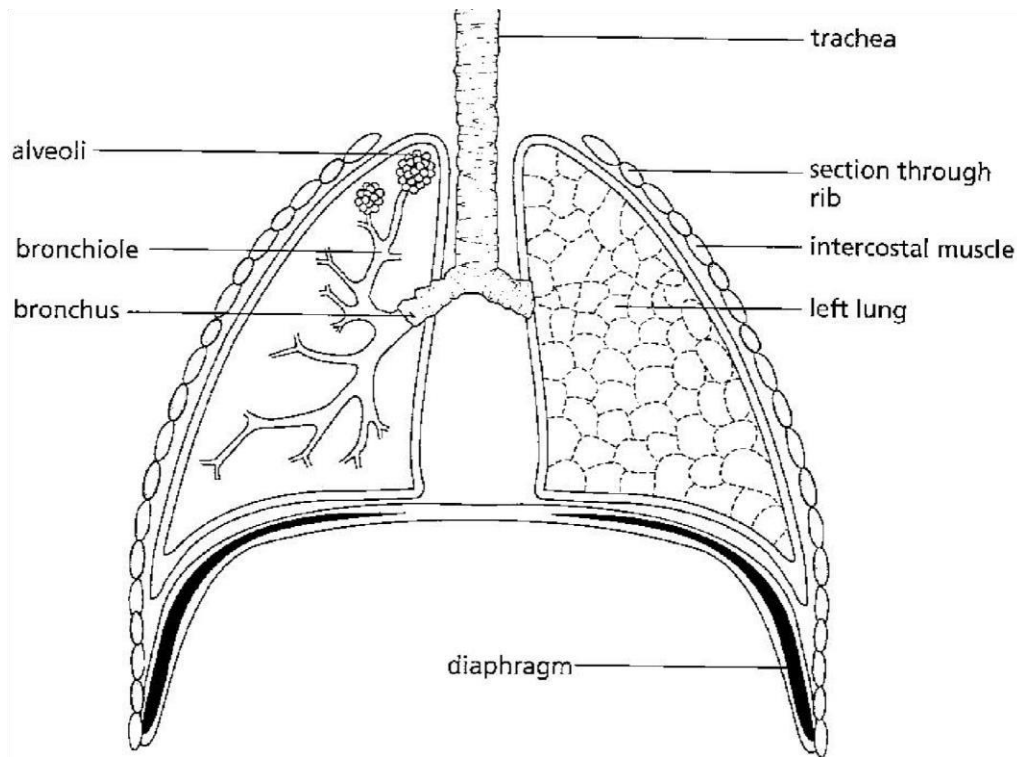
- It is a diet that includes all the essential nutrients in their correct proportions to keep one healthy

Components of a balanced diet

Component	Functions of nutrients	Food source
Carbohydrates	- Provide the body with energy	Bread, maize, yams, sweet potatoes, rice
Fats	- Provide energy - Store energy in the body - Reduce heat loss through the skin	Meat, butter, oil, milk, cheese, yoghurt, nuts
	- Protect organs in the body from physical damage	
Proteins	- Needed for growth, especially of muscles and nerves - Needed to repair damaged tissues	Meat, chicken, fish, eggs, beans, milk products
Vitamins	- Important for fighting infections - Improves wound healing - Prevents blindness	Fruits, vegetables, milk, eggs
Mineral salts	- Needed for formation of strong bones and teeth - Needed for blood formation - Needed for growth and development	Vegetables, meat, milk, nuts
Fibre	- Assists digestion - Needed for normal bowel movement - Prevents constipation	Fruit, vegetables, wholegrain foods
Water	- Enables chemical reactions to take place in the body. - Transport substances around the body - Cools the body	Water melons

RESPIRATORY SYSTEM

RESPIRATORY ORGANS



- Air is taken through the nostrils and then travels via the trachea into the right and left bronchi, which divides and re-branches into bronchioles, each of which ends in a cluster of alveoli. It is just in the alveoli that the actual gaseous exchange occurs
- Parts of respiratory system

Nose and nasal passages

- The nostrils contains hairs that assist in filtering out dirt from entering air
- The nasal passages warm the air and add moisture to the air. Mucus in the nasal passages traps bacteria and dust.

Mucus is antiseptic i.e. kills bacteria. Trachea (wind pipe), bronchi and bronchioles

- Walls of the trachea are supported by c-shaped rings of cartilage. This prevents the trachea from collapsing and thus holds it open for free passage of air.
- The trachea divides into bronchi. Each bronchus extends to each lung and branches into numerous small bronchioles which further channels air to and from the alveoli.
- Particles of dust and bacteria are trapped in the film of mucus covering these cells
- The trachea and its branches are also lined with cilia which are in constant motion and carry dust and dirt mixed with mucus, upwards, towards the mouth.

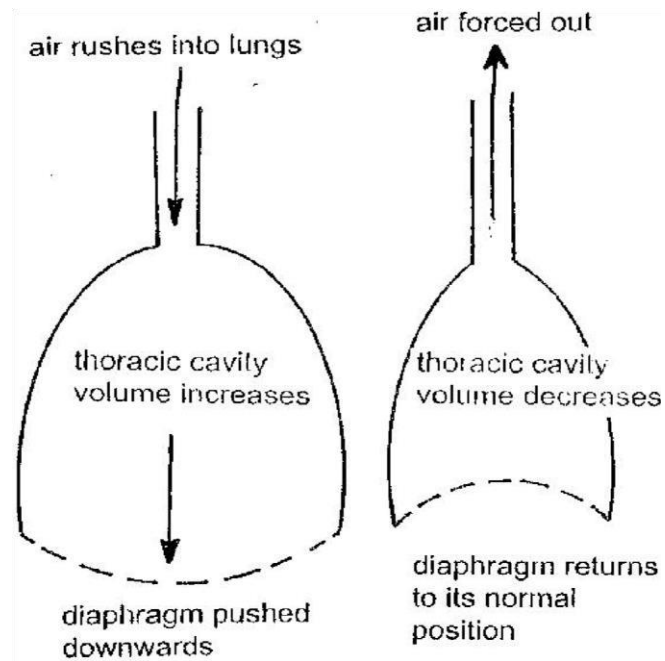
The air sacs

- The bronchioles branch repeatedly into smaller and smaller passageways which end in air sacs (alveoli)
- These are the respiratory surfaces where gaseous exchange takes place
- The air sacs are thin walled and covered by numerous blood capillaries to facilitate efficient gaseous exchange.

Breathing mechanism

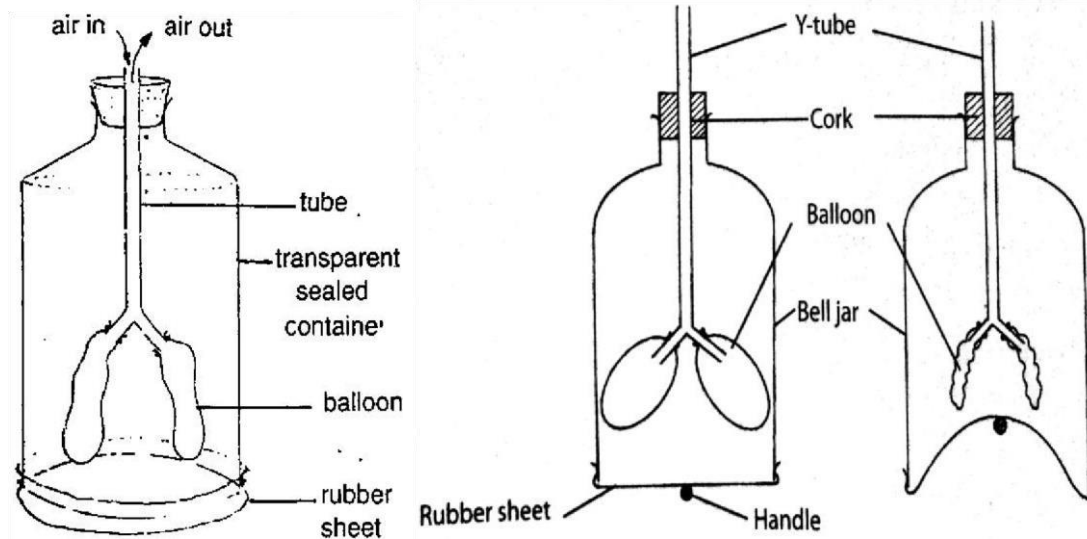
- Breathing is the ventilation of lungs by the movement of ribs and diaphragm causing air to enter lungs to bring oxygen and leave lungs to remove carbon dioxide
- Breathing involves two sets of muscles i.e. diaphragm and intercostal muscles which work to increase or decrease the volume

of the thoracic cavity, so that pressure is increased or decreased and air moves in and out of lungs



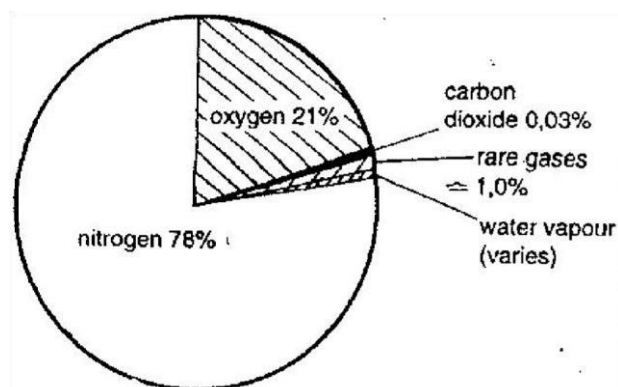
- During breathing in (inhaling);
 - The external intercostal muscle contract while internal intercostal muscles relax, lifting the rib cage upwards and outwards. The diaphragm muscle contracts and moving the diaphragm down (flattens). The thoracic volume increases and pressure of air in the thoracic cavity increases. Air moves into the lungs.
- During breathing out (exhaling);
 - The external intercostal muscles relax while internal intercostal contract, lowering the ribcage downwards and inwards. The diaphragm muscle relaxes and the diaphragm arches upwards. Thoracic volume decreases and pressure is increased. Air is forced out the lungs.

Demonstrating breathing



- A piece of apparatus called bell - jar model can be used to show the way in which movement of the diaphragm results in inspiration and expiration.
- The balloon starts off deflated. When the rubber sheet is pulled down, the balloons inflate, if released the balloon deflate again.
- When the rubber sheet is pulled down, the volume inside the bell jar increases. This reduces the air inside, making it lower than the outside. The air rushes in through the glass tubing, to equalise the air pressure, causing the balloons to inflate

The composition of air



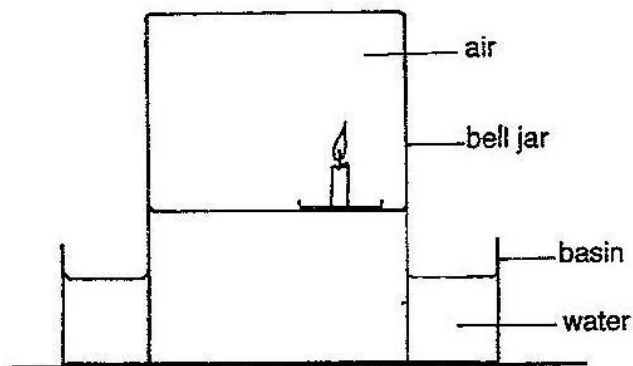
- Air is a mixture of gases i.e. nitrogen, oxygen, carbon dioxide, rare gases and water

- Air we breathe in contains about 21% Oxygen, some is used by cells during respiration resulting in reduction of oxygen in exhaled air to 16%
- The remaining 79% of the air consist mainly of nitrogen, the percentage composition of which does not change significantly during breathing.
- Inspiration air contains 0.04% carbon dioxide. Cells of the body produce carbon dioxide as a waste product during respiration. Expired air contains 4% carbon dioxide

Respiratory gases

- Respiratory gases are gases in air that are involved in respiration. These include;
 - ✓ Oxygen
 - ✓ Carbon dioxide
 - ✓ Water vapour

Experiment to compare oxygen in inhaled and exhaled air



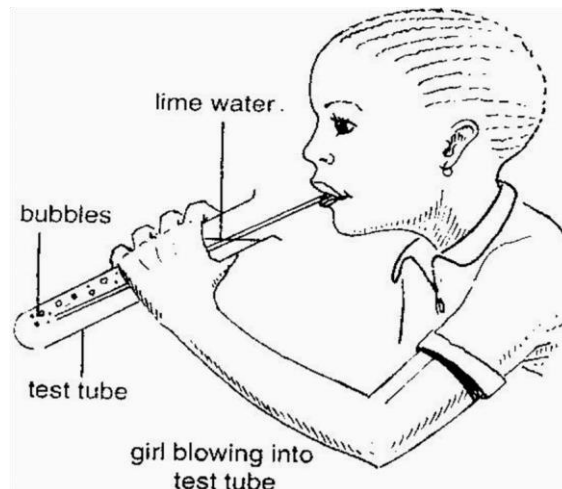
- Place a candle inside a container with inhaled and exhaled air and measure the water displacement.
- Water displacement in inhaled air is greater and the candle burns for a longer time than in exhaled air. This is

because inhaled air contains more oxygen and less carbon dioxide when compared to exhaled air which contains less oxygen and more carbon dioxide.

- When candles were burning they used up oxygen. Water rise is highest in jar with most amount of oxygen

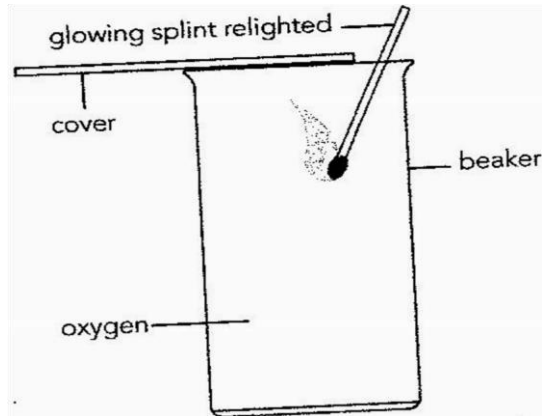
The test for carbon dioxide

- Blow air into limewater or bicarbonate indicator.
- Limewater changes from clear to milky and bicarbonate indicator changes from red to yellow.



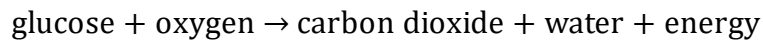
The test for oxygen

- Insert a glowing wooden splint into a test tube with oxygen.
- The glowing splint rekindles or relights in the presence of oxygen.



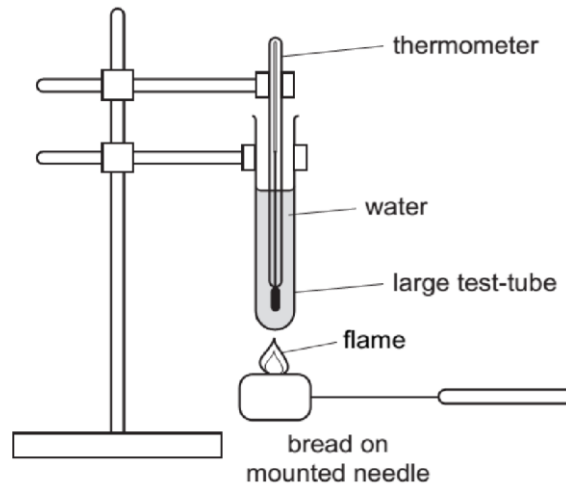
RESPIRATION

- It is the breakdown of carbohydrates to release energy



Experiment to show that energy is released during burning of during respiration

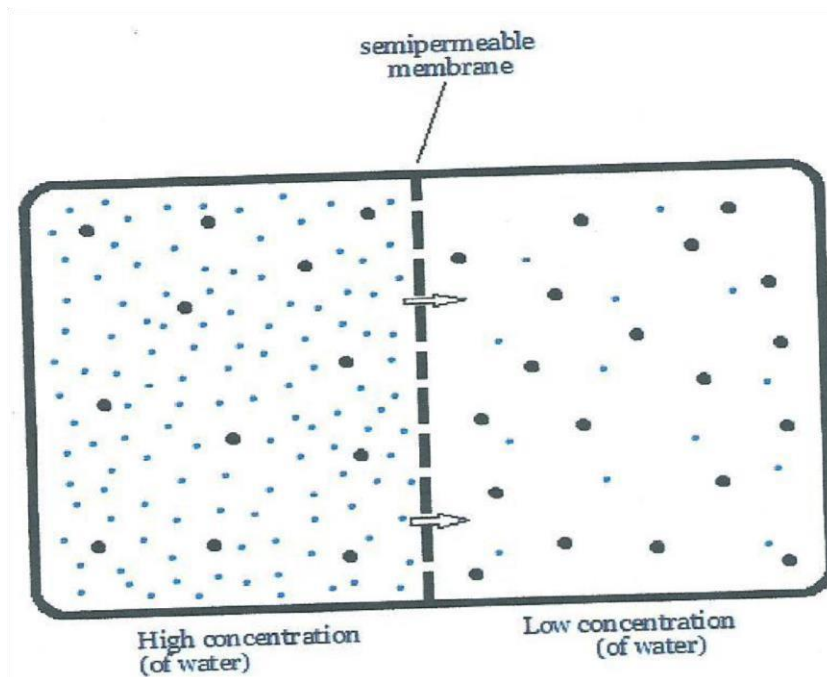
- Place some mealie-meal in a lid or crucible and heat the meal until it begins to burn.
- Light the meal in the dish and remove the burner so that the burn on its own.
- Place a small container of water above the burning meal
- Record the temperature of the water before and after the meal has burnt away.
- The temperature increase is caused by the heat given off by the burning mealie meal. The meal had chemical energy stored in it.
- The energy trapped in plant material by photosynthesis is released in the body by respiration.



TRANSPORT SYSTEM

TRANSPORT IN PLANTS

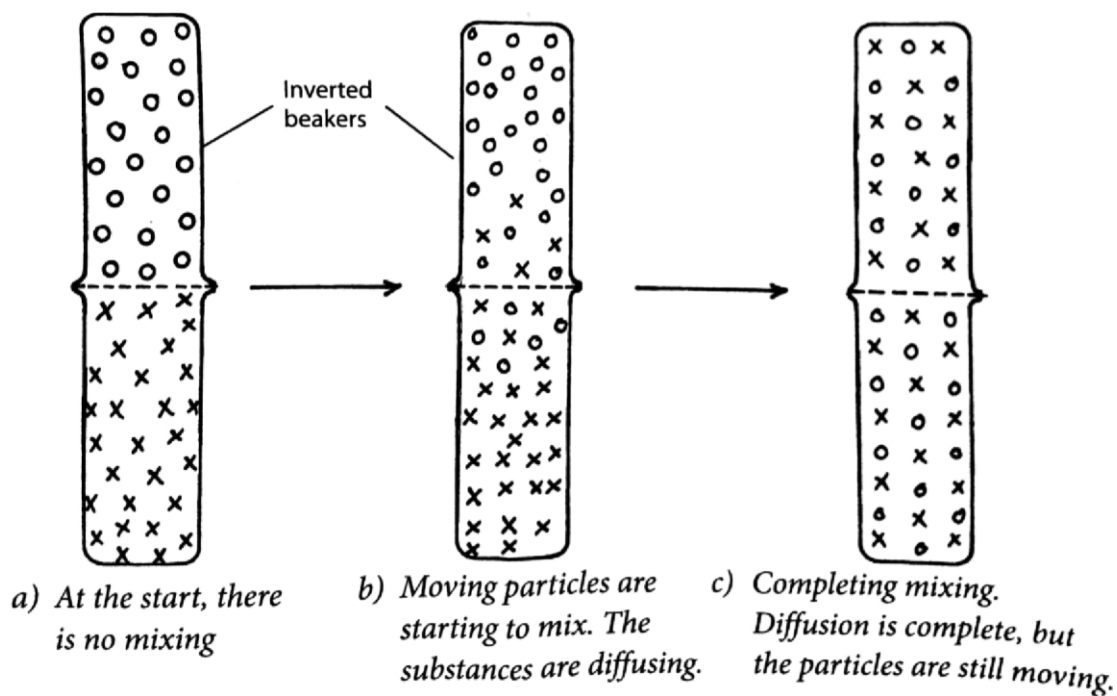
OSMOSIS



- Osmosis is the net movement of water molecules through a partially (semi) permeable membrane from an area of higher water concentration.
- A partially permeable membrane allow small molecules (water) to pass through them and prevented the larger molecules from passing through until the water particles are evenly spread out i.e. at equilibrium

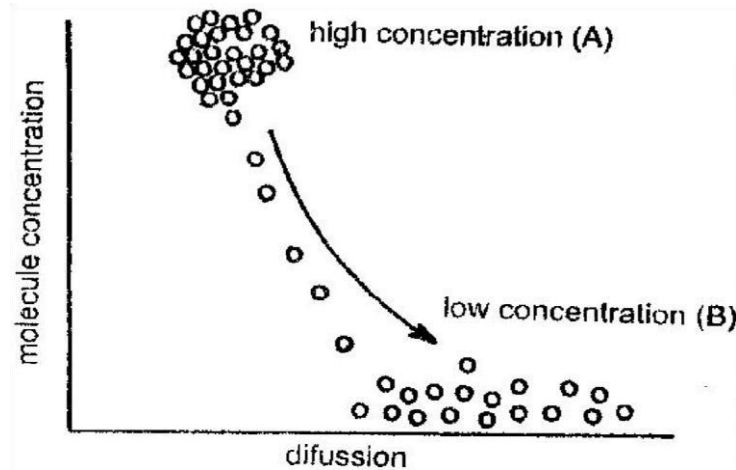
Diffusion

- Diffusion is the net movement of particles from a region of their concentration to a region of their lower concentration down a concentration gradient, as a result of their random movement

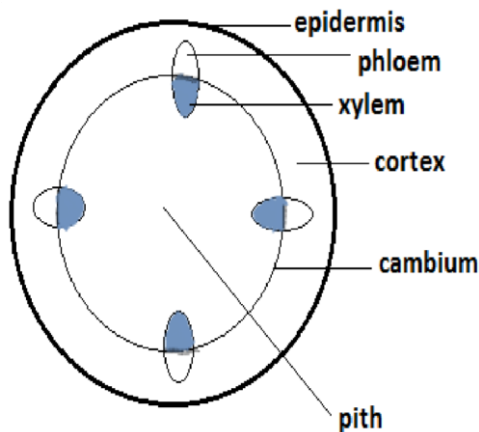
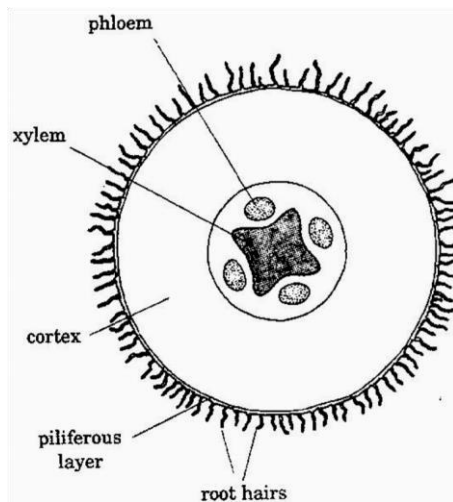


- Diffusion will always proceed whenever there are differences in concentration until the molecules are evenly distributed throughout the system
- The bigger the difference in concentration of a substance, the faster it will diffuse

- Molecules are in random motion but there is a net movement of molecules away from the more concentrated area.



ROOT AND STEM STRUCTURE Internal structures of a dicotyledonous root and stem



Epidermis

For absorption i.e. the cells have large central vacuoles and large surface areas for the uptake of water through osmosis and protect the inner root tissue.

Cortex

Provide a pathway for water to the xylem, but can also store water and food.

Phloem

Conduction of food materials (translocation).

Xylem

Conduction of water and dissolved salts and support.

Cambium

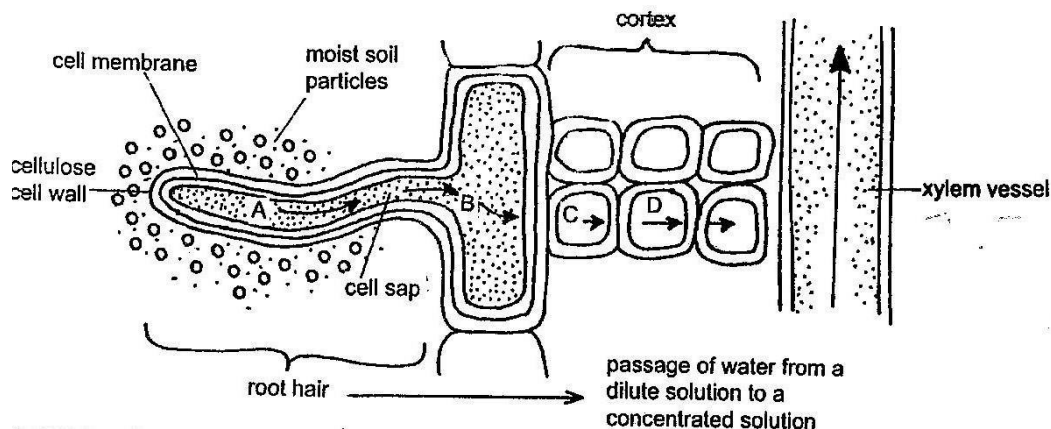
Formation of secondary xylem and phloem resulting in growth in

diameter.

Root hairs they provide the main absorbing region for water and mineral salts from the soil. They are numerous to provide more surface area for absorption.

Water and ion uptake

- Water is drawn into the root hairs mainly by osmosis.
- The cell sap in the root hair cells is more concentrated with sugars and salts than the soil water. Water molecules are drawn across the permeable cellulose cell wall and semi-permeable membrane and protoplasm then into the vacuole.
- After water has been absorbed into the root, there are more water molecules in region A than B, so water diffuses from A to B then C and D until it reaches the xylem.
- Molecules or ions move from a region of low concentration to higher concentration
i.e. they move against concentration gradient and energy is used to move molecules from the soil water into the root hair cells. This kind of movement of molecules is called active transport.
- The use of energy enables ions to move against the concentration gradient.

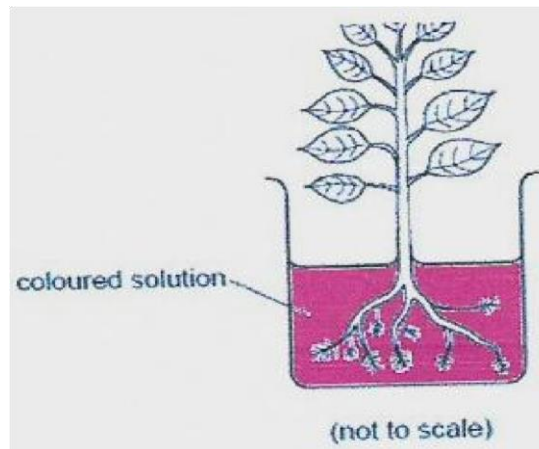
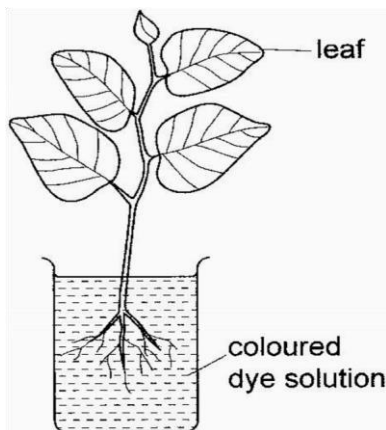


Water movement in plants

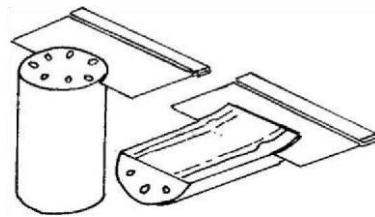
- The xylem vessels conduct water to the leaves where it diffuses to all cells.
- Water vapour evaporating from a leaf creates a kind of suction drawing a stream of water upwards, as water molecules are attracted to each other. This creates a transpiration stream, pulling water up from the root.
- Xylem vessels act like fine tubes drawing water up the stem by capillary action. Water molecules have high surface tension and hold to each other well enough to pull each other up.
- Roots also produce a root pressure i.e. the water absorbed osmotically by the roots, force water already present in the xylem upwards in the stem

Demonstrating water movement in a plant

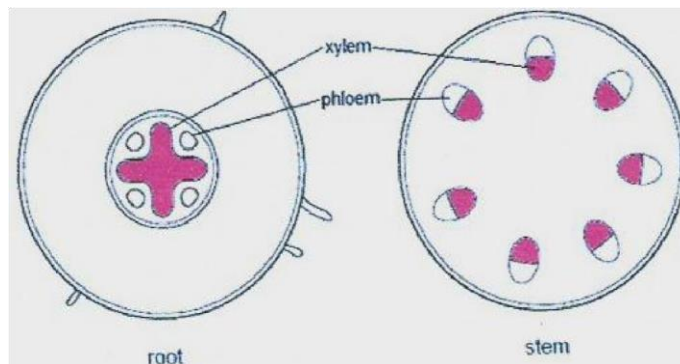
- To track the movement of water in a plant, a coloured solution can be used



- A young dicotyledonous plant was uprooted, placed in a beaker with coloured water, left in sunny place for a few hours
- Cut off thin slices of plant roots and stem to trace the movement of the dye.



- Examine these slices using a hand lens
- Make simple diagrams to show the distribution of the dye.
The vascular bundles (xylem) will have been stained by the colouring since water moved by osmosis up the stalk.

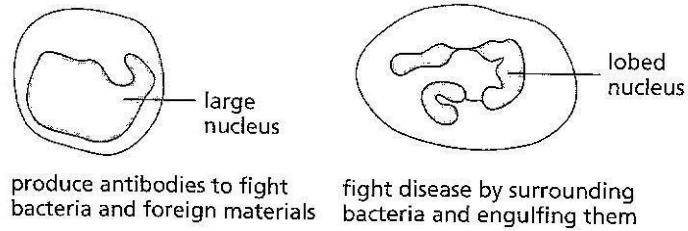


- The dye stain distribution pattern shows the pattern of xylem and hence vascular bundle distribution in the dicotyledonous stem and root.
- Water is drawn up in the plant through the xylem vessels which are part of the vascular bundle.

TRANSPORT IN HUMANS

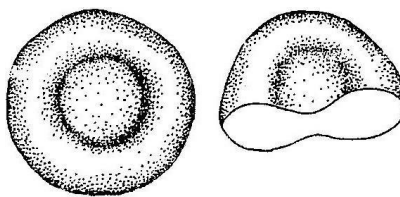
CIRCULATORY SYSTEM Blood components

White blood cells (leucocytes)



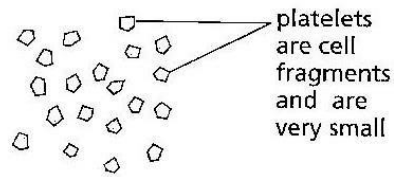
- Destroy bacteria and fight infection through phagocytosis and production of antibodies. They protect the body from an infection.
- Phagocytes engulf invading micro-organism by a process called phagocytosis i.e. they ingest and destroy pathogens to prevent infection.
- Lymphocytes produce antitoxins (antibodies) which neutralize the toxins produced by bacteria or agglutinate pathogens

Red blood cells (erythrocytes)



- Are small disc-like cell which don't have a nucleus. They contain hemoglobin, red pigment which combines with oxygen and is taken to all body tissues thus red blood cells transport oxygen.

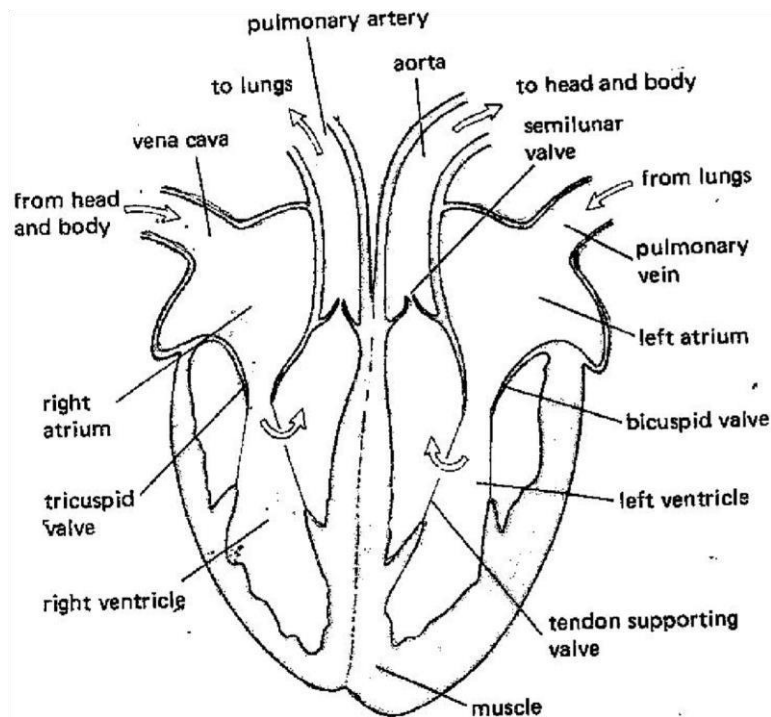
Platelets (thrombocytes)



form blood clots, which stop blood loss at a wound and prevent the entry of germs into the body

- Platelets are small structures which assist in blood clotting.
- Whenever a blood vessel is damaged, platelets secrete an enzyme (thrombokinase) which act on a blood protein (prothrombin) and turns it into thrombin. This in turn assist in the conversion of soluble fibrinogen in to insoluble fibrin which forms an obstruction in which the blood cells become entangled thus stopping bleeding.
- The clot helps to prevent disease causing organisms from entering the body. Plasma
- Plasma is a yellowish and slightly alkaline liquid part of the blood which carries the majority of the transport function.
- Plasma contains dissolved substance like blood proteins (fibrinogen), inorganic salts of (Ca, Na & K), organic constituents (glucose & urea) dissolved gases (oxygen & carbon dioxide) and secretions (hormones and enzymes)
- Blood plasma is a medium in which blood cells and platelets move to all parts of the body.

The heart



Functions of the heart

- The heart pumps and receives blood to and from all body parts.
- The heart is a muscular organ made of cardiac muscles that contract rhythmically to enable the heart to pump blood out.
- It consists of two atria – receiving chambers and two ventricles – pumping chambers.
- The septum separates the left and right sides preventing mixing of oxygenated and deoxygenated blood.
- Walls of the left ventricles are three times thicker than the right ventricle to generate enough pressure to move the blood to all organs of the body. Right ventricles only pumps blood to the lungs.

- Atrium opens into corresponding ventricles. Valves allow blood to flow in one direction or prevent back flow of blood.

Blood circulation

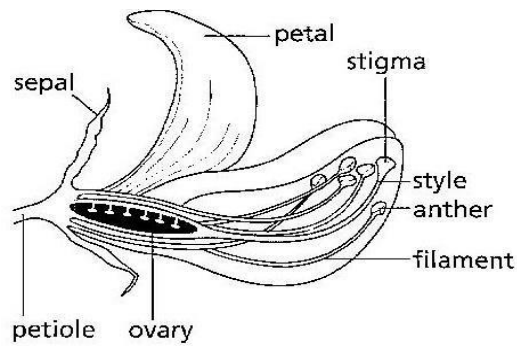
- Blood in the right ventricle is pumped to the lungs. Blood from the lungs flows back into the left atrium through the pulmonary vein and then is pumped into the left ventricle
- Blood in the left ventricle is pumped to the rest of the body through the aorta. Blood returns to the heart where it enters the right atrium through the vena cava. The right atrium pumps blood to the right ventricle. The main blood vessels to and from the heart
- Vena cava - carries deoxygenated blood to the heart
- Aorta - carry deoxygenated blood from the heart to the body
- Pulmonary artery - carry deoxygenated blood from the heart and lungs
- Pulmonary vein - carry oxygenated blood from lungs to the heart

REPRODUCTION

REPRODUCTION IN PLANTS

The structure of a flower

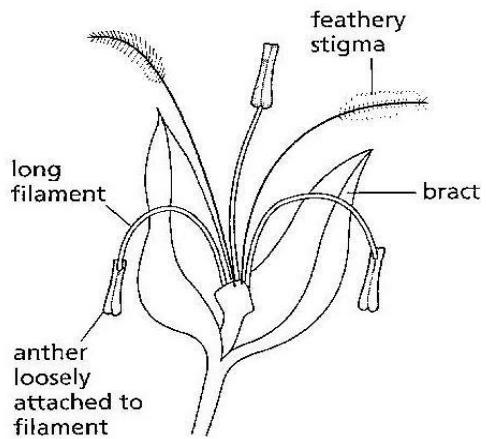
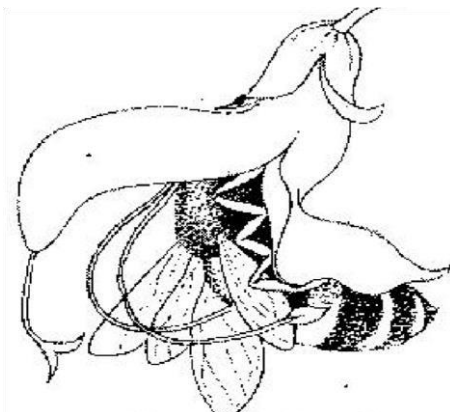
- A flower is the reproductive organ of plants. It consists of carpel (female parts) and the stamens (male parts).



1. Stamens - consist of the anther and the filament. The anther produces pollen grains which carry the male sex cells. The filament holds and supports the anthers in position
 2. Carpel/pistils - consist of a sticky top called stigma where pollen grains are deposited during pollination, a slender stalk (style) which supports the stigma and is the pathway for pollen tubes to ovules, and a swollen base (ovary) which contains ovules which carry the female sex cells.
 3. Filament - hold and support the flower
 4. Sepal - protects the flower when it is in bud
 5. Petal - are often, but not always, large, scented and coloured.
 6. Receptacle - supports the floral whorls
- Pollination
- Pollination is the physical transfer of pollen grains from the anther to the stigma of the same species.
 - It is important because it carries the male sex cells in the pollen grains to the female part of the flower
 - There are two types of pollination i.e.

1. Self pollination – takes place when pollen grains are transferred from the anther to the stigma of flowers within the same plant.
2. Cross pollination – takes place when pollen grains are transferred to flowers from different plants but of the same species.

- Pollination is aided by insects or wind or birds

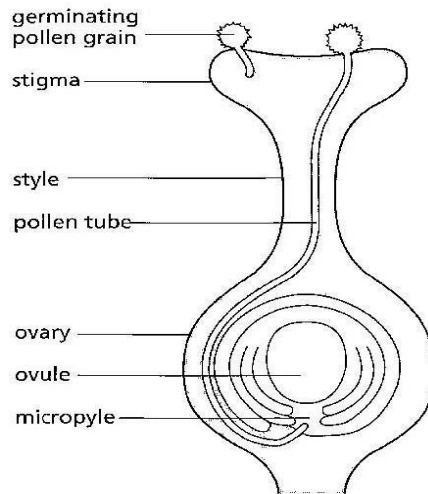
Wind pollinated flower	Insect pollinated flowers
 The diagram illustrates a wind-pollinated flower. It features a long, thin filament supporting a feathery stigma at the top. The anther is shown loosely attached to the base of the filament. A bract is also visible near the base of the flower.	 The illustration shows an insect pollinated flower. A bee is depicted hovering near the flower, which has a prominent, curved structure (likely a nectar guide or landing platform) and a large, dark, patterned area (likely the corolla or a nectar source). The bee is positioned to collect nectar or pollen from the flower.

- Abundant, dry or light pollen grains that can be carried by the wind - Petals are small to expose stamens and carpel to the wind. - Petals are dull (green) - Anthers and stigmas are long so they are exposed to the wind. - Stamens are exposed to the wind so that the pollen can be carried away by the wind. - Stigma feathery to provide large surface to trap as much pollen as possible from the wind.	- Have large coloured and scented petals to attract insects and birds. - Nectary produces sweet nectar which attracts insects. - Stigmas and anthers are enclosed inside the flower protected by petals. - Anthers have sticky pollen grains that stick to the insect's body. - Stigma is also sticky so that pollen grains on the insect's body stick to the stigma.
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Fertilisation

- Fertilisation is the fusion of male and female sex cells to form a zygote.
- When pollen grain is deposited on the stigma, it germinates a pollen tube which grows down the style carrying the male nucleus along with it.
- Upon reaching the ovule, the tube breaks open and the male nucleus fuses with the female nucleus to form a zygote
- After fertilisation the following changes occur;
 1. Fertilised ovule divides by mitosis to form a seed containing the embryo plant (plumule & radicle) and food stores (cotyledons or endosperms)

2. The wall of the ovule forms the seed testa (coat). The ovary walls develop into a fruit, which may be fleshy or dry pod.
3. All other parts shrivel and die

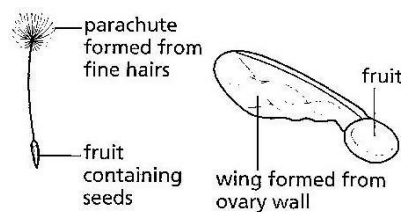


SEED DISPERSAL

- Seed dispersal is the scattering of seeds far away from the plant in order for new plants to stand a good chance of growing to maturity.

Wind dispersal

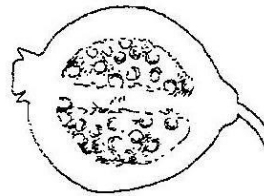
- Fruits usually have a wing or hairs to help them be carried by the wind from the parent plant e.g. sycamore and dandelion



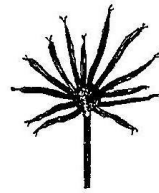
Animal dispersal

- There are two main modifications of fruits for animal dispersal i.e.

- Succulent fruits which attract animals because they have juicy flesh which is pleasant to eat but the seed is not palatable, so it is deposited in animals' droppings some distance from the parent plant e.g. tomatoes, guavas

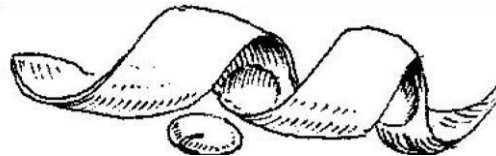
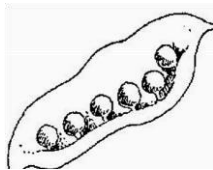


- Hooked fruits which catch or cling on to an animal's fur as it brushes past the parent plant. Eventually the seeds drop off, a further distance from the parent plant e.g. black jack



Mechanical (self) dispersal

- Pods often twist as they ripe because of changes in the amount of moisture in the air causing strains on the pod. The pod burst open suddenly and with enough force to throw its seeds some distance from the parent plant e.g. beans, peas

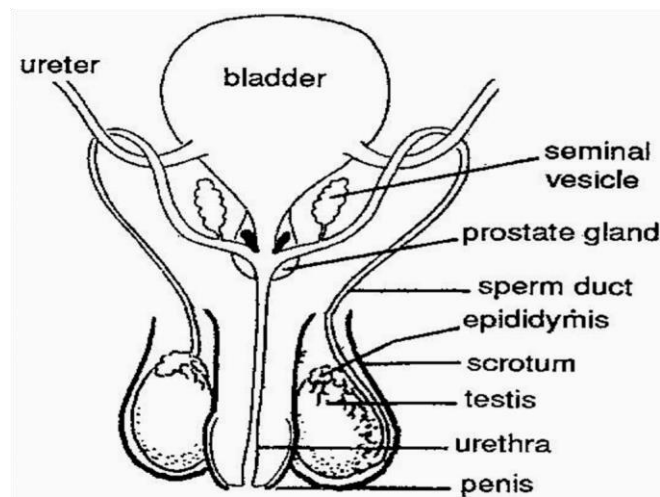


Water

- Fruits are light such that they float away and may eventually be washed on riversides where the seed develops into a new tree e.g. coconut.

REPRODUCTION IN HUMANS

Male reproductive system



Testes Is the male gonad that makes and produce sperm cells. Produce male hormones which cause secondary sexual characteristics in males.

Scrotum It is a sac that holds testes outside the body keeping them cooler than body temperature to favour sperm production.

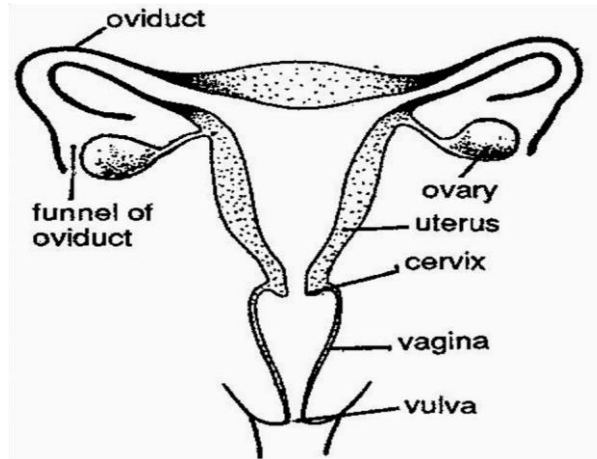
Epididymis Mass of tubes in which sperms are temporarily stored.

Sperm duct Is a muscular tube that links the testes to the urethra to allow the passage of semen containing sperms

Urethra Serves as a duct for passage of semen and urine through the penis.

Penis Consist of erectile and connective tissue with numerous blood vessels. During excitation, it becomes engorged with blood, stiffens and become erect, inserted into the vagina in order to transfer sperms.

Female reproductive system



Ovaries Contain follicles in which ova or eggs are produced. They also produce the female hormones which causes secondary sexual characteristics in females

Oviduct Carries an ovum to the uterus, with propulsion provided by tiny cilia in the walls. This is also where fertilisation takes place.

Uterus Is where the fertilised ovum develops into a foetus during pregnancy

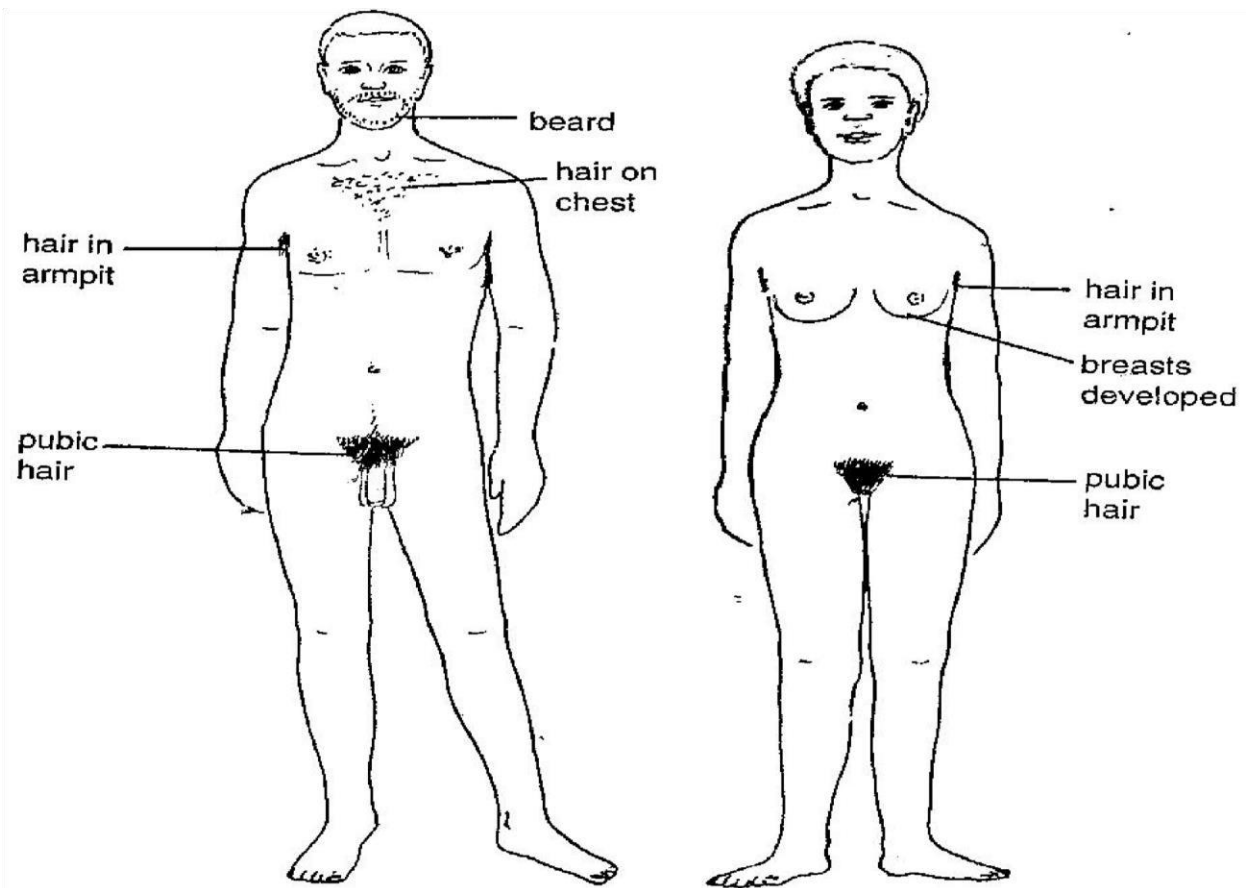
Cervix Is a narrow ring of muscle that closes the uterus but can expand greatly during birth of a baby.

Vagina Is a muscular organ which opens to the outside of the body. The vagina receive male penis during copulation and it is also where sperms are deposited during copulation.

PUBERTY

- Puberty is the stage at which sexual organs mature for reproduction.
- In males it starts between the age of 9 and 14 while in females it starts between

813 years



Signs of puberty

Boys

- Begins at around the age of 12 - 14 years. The signs include;
 - ✓ Body hairs grow e.g. in arm pits, on chest, face and pubic regions
 - ✓ Lengthening and widening of the penis
 - ✓ Increased mass and height
 - ✓ Deepening of voice due to enlarged voice box
 - ✓ Broadening of shoulders

- ✓ Wet dreams occurs i.e. involuntary ejaculation of semen during sleep

Girls

- Begin at the age of about 10 to 13 years. Changes are mainly caused by the hormones oestrogen and progesterone. The signs include;
 - ✓ Onset of menstruation
 - ✓ Widening of the pelvic girdle (hips) to prepare the body for carrying and delivering a baby.
 - ✓ Enlarged breasts
 - ✓ Body hair grows i.e. in armpits and pubic regions - Premenstrual symptoms include;
 - ✓ Mood swings and irritability or anger
 - ✓ Poor concentration
 - ✓ Social withdrawal
 - ✓ Change in libido
 - ✓ Tension and anxiety
 - ✓ Breast tender
 - ✓ Nausea and constipation or diarrhea
 - ✓ Period pain when the uterus contracts
 - ✓ Cramps
 - ✓ Acne

HEALTH AND DISEASES

HEALTH

- Health is complete state of physical, mental and social well being. Healthy is the well being of the body in its entirety i.e. with all the normal functions well coordinated.
- Mental well being is to do with a person's state of mind and social well being is to do with having decent accommodation, clothing, food and being part of a functional community and physical well being is to do with how well your body functions
- To enjoy good health, a person needs proper sleep and rest, proper shelter, balanced diet and enough exercises or physical fitness

DISEASES

- A disease can be defined as a disorder or malfunction of the body which leads to a departure from good health
Causes of disease
- Poor nutrition e.g. kwashiorkor
- Pathogens e.g. bilharzia (worms), malaria (protozoa), polio (virus) and typhoid (bacteria)
- Chemicals and poison e.g. lung cancer, asbestosis, liver cirrhosis, emphysema
- Genetic defects e.g. albinism, anaemia, colour blindness, Down's syndrome, dwarfism

Methods of transmission of diseases

- Disease transmission is the way in which a disease is spread. These include; Contact
- Pathogens can be passed from one person to another when people are in close contact, directly or indirectly
- Direct requires close physical contact between an infected person and a susceptible person and the physical transfer of micro-organisms. This includes touching an infected person, kissing, and sexual contact, contact with oral secretions or body lesions. Diseases include Ebola, STIs etc
- Indirect contact transmission refers to situations where a susceptible person is infected from contact with contaminated surfaces e.g. clothes. Diseases include ring worm and Tinea, Ebola etc

Droplet infection

- Refer to situations where droplet nuclei or dust containing micro-organisms can remain suspended in air for long periods of time.
- Airborne transmission allows organisms to enter the upper and lower respiratory tracts
- Some diseases are transferred by infected droplets contacting surfaces of the eye, nose, or mouth.
- Droplets containing micro-organisms can be generated when an infected person coughs or sneezes. E.g. TB, typhoid, influenza, polio, pneumonia, whooping cough

Vectors

- Are organisms capable of transmitting diseases e.g. mosquito, flies or ticks etc.
- Vectors feed by biting through the skin and then sucking blood. If there were any pathogens in the saliva of the insect then these would be injected into the blood of the host animals.
- Micro-organisms could also be located on the outside surface of a vector or their faeces and spread through physical contact with food, a common touch surface, or a susceptible individual.
- Diseases include malaria, cholera Contaminated water
- Water is a favorable medium for the dispersal of the organisms causing gastrointestinal infections such as dysentery, cholera and typhoid.
- When the organisms reproduce in the gut, eggs are carried out with the faeces. If the faeces or urine carrying pathogens are deposited directly in water that is used for drinking, the organisms may thus infect large number of people.
- Disease organisms are washed into water supply and contaminate it.

Contaminated food

- Food may be contaminated with pathogenic organisms in a number of ways i.e. contaminated hands, flies and infested

meat or washing in contaminated water. - Diseases

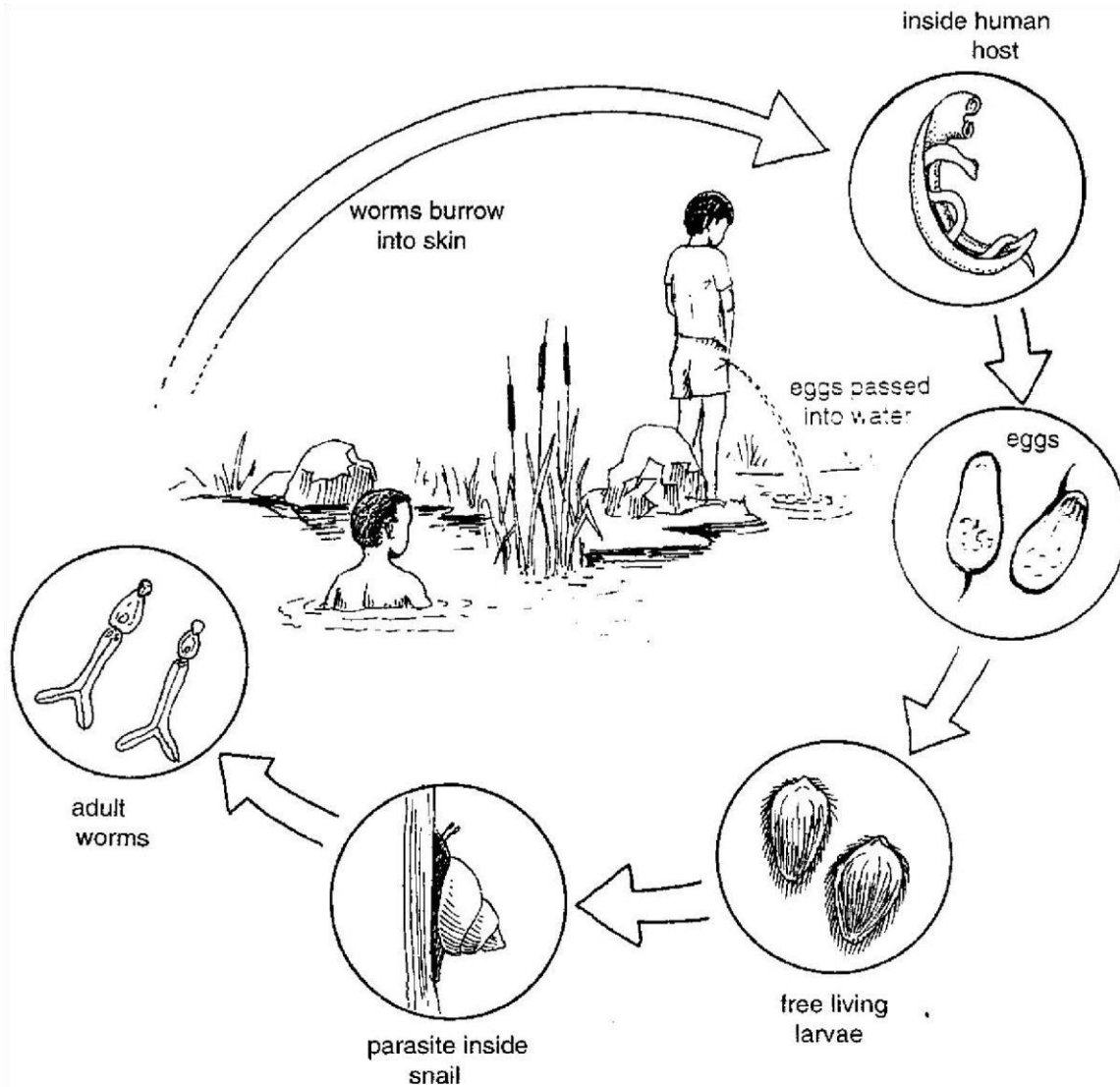
include typhoid, dysentery

BILHARZIA

Causes of Bilharzia

- Is caused by parasitic flat worm (blood fluke) called *Schistosoma* which invade blood vessels of the gut and the walls of the gut and walls of the urinary bladder.

Bilharzia Life Cycle



- The parasite's life cycle occurs in fresh water snail and a human
- Once flukes enter the body, they feed mainly on blood and tissue directly from the wall of the urinary bladder, to which they attach themselves by suckers.
- The flukes reproduce sexually and eggs are laid in the blood vessels of the gut or bladder.

- Each egg has a conspicuous spine and, as the embryo develops it vibrates, the spine cuts a hole in the wall of blood vessels. The eggs with a little blood enter the intestine or the bladder, to be dispersed either in faeces or in urine.
- The eggs will then hatch and develop into a series of different larvae inside the water snail. The larvae reproduce themselves asexually in the snail.
- The parasite leaves the snail and enters the water. It now has a forked tail to help it to swim.
- The larvae are able to penetrate a person who is bare footed in stagnant water or slow moving water bodies and immediately penetrate and invade the blood vessels. The larvae may also penetrate the body from the gut if they are swallowed in untreated drinking water.

Control or prevention

1. Avoid standing in water containing snails
2. Eliminate water snails by molluscides, which clear weed on which snails feed
3. Filtration and chlorination or boiling of water before use
4. Avoid urinating or defecating in water bodies
5. Use of drugs e.g. oxamiquine

HYGIENE

- Hygiene refers to conditions and practice that prevent the spread of disease.

- Good personal and food hygiene are important to prevent the spread of diseases.

Personal hygiene

- Involves taking care of your own body i.e.
 - Bathing daily using soap and water
 - Washing hands and scrubbing nails before preparation of food or eating
 - Clean the pubic and anal region
 - Brushing teeth twice a day
 - Washing and ironing clothes frequently
- Such simple precautions greatly reduce the chances of picking up an intestinal disease and also prevent passing on of diseases
- It also helps prevent fungal infections and will remove body parasites which might be carrying disease organisms.

Food hygiene

- Involves handling, serving, storing and preparation of food in a way that prevents the spread of diseases.
- It is important that a person who is working with food washes his/her hands with soap and water before preparing the food.
- Food should be stored at correct temperature and should be cooked correctly and thoroughly to kill bacteria. Food that has passed its expiry date should not be eaten

Toilet hygiene

- The toilet is a place where bacteria can be found in the toilet bowl, on the seat, under the lid, on the handle and on the floor.
- A firm bristled toilet brush is usually used to clean any waste and disinfectants are used to kill bacteria in the toilet.

WASTE DISPOSAL

- Litter is rubbish such as paper, tins and bottles that are left lying around.
- Wastes refer to any material that is unwanted i.e. domestic or industrial
- Some wastes can be naturally broken down by organisms and are called biodegradable waste. Some cannot be broken down naturally and are called non biodegradable wastes
- If wastes are not disposed of properly, it creates a health hazard. The wastes attract flies, mosquito and rats that spread diseases
- All wastes need to be removed or disposed of. Methods of disposal include;

Burning

- Wastes are burnt in bins or incinerators
- Advantages
- Reduces the volume of wastes
 - Less expensive

- Prevents contamination of ground water by chemicals
- Pathogens destroyed
- Easy and takes less time (fast) Disadvantages
- Causes air pollution
- Produce harmful chemicals
- Time consuming to separate wastes that can not be burnt
- Leads to greenhouse effect and hence global warming - Expensive to

control emissions into the air

Burying

- Waste is buried in a managed landfill sites. Advantages
- Cheapest form of solid waste disposal
- Filled land can be used for other purposes
- Can handle large amount of wastes
- Prevents bad odour that emanate from decomposing wastes
- Prevents insects and rodents from breeding in the wastes and cannot spread diseases

Disadvantages

- Requires large areas of land
- Contaminates the soil and or groundwater
- Poorly managed harmful substances create a serious health risk to humans and animals

- Emits greenhouse gases
- Can attract disease carrying animal

Recycling

- Recycling means to use waste materials to make other products. Waste paper, plastics, metal and glass can be recycled and remoulded into the same or other manufactured goods. Wastes for recycling are usually sorted in recycling bins. Advantages

- Reduces the amount of wastes that needs to be disposed of.
- Conserves natural resources, since there is less demand for raw materials
- Creates income or employment
- Reduces pollution

Disadvantages

- Could be unhygienic
- Sorting can be time consuming and expensive

COMBINED SCIENCE
CHEMISTRY SECTION

2019

DISCOVERING COMBINED SCIENCE BOOK 1-2



TOPICS

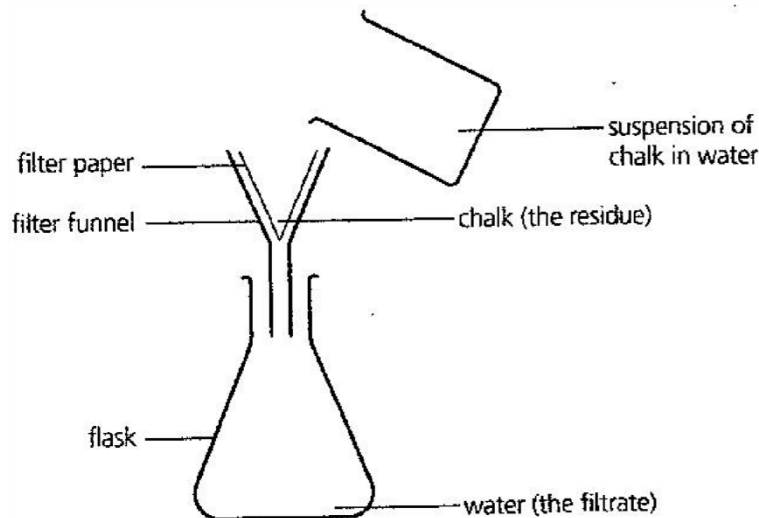
- SEPARATION
- MATTER
- ACIDS, BASES AND SALTS
- OXIDATION AND REDUCTION
- INDUSTRIAL PROCESSES
- ORGANIC CHEMISTRY

SEPARATION OF MIXTURES

METHODS OF SEPARATING MIXTURES

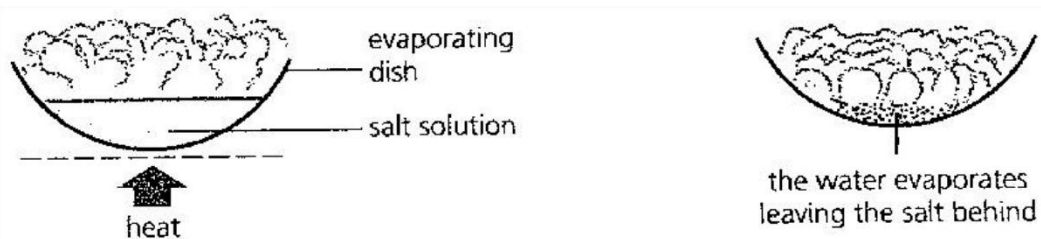
Filtration

- It is a process of passing a fluid through a filter to remove solid particles.



- Is used to separate insoluble solids from liquids e.g. a mixture of soil and water.
- A filter is placed in a funnel, on a flask and the mixture is poured on the filter paper. The liquid filtrate passes through the filter paper and is collected. Insoluble substances are left as residue on the filter paper after filtration.
- Filtration can be used in the following process
 - Water treatment i.e. filtration removes all suspended solids like plants and animal matter from water before it is made available for use.

Evaporation



Is used to separate a solution in which the solid is dissolved in the liquid.

- The solution is heated so that the solvent evaporates and the solid is left behind after all the liquid has evaporated.
- Evaporation is used in the following processes
 - Formation of sugar crystals – evaporation plays an important role in the extraction of crystals of sugar from sugar cane juice. The clear juice is piped into evaporators which reduce the mixture to thick sugary syrup.
 - Formation of ammonium nitrate crystals from their solutions. Excess water in the ammonium nitrate solution is evaporated to produce a saturated solution. The ammonium nitrate crystallizes out of the solution.
 - Table salt processing – shallow ponds of sea water are left to dry in sunshine until all water has evaporated.
 - Salting peanuts – g

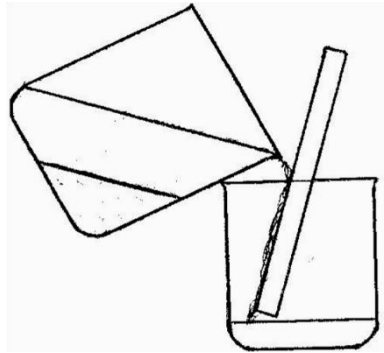
Winnowing



- Winnowing is used to separate solids with different densities. Lighter materials will blow away while the heavier material will fall to the ground.
- A winnowing basket is shaken vigorously from side to side. The less dense come to the top and are blown off while the heavier remain in the container - Winnowing can be used in the following processes
 - Grain separation – winnowing by wind is a method of separating usable grain from husk (chaff). To remove unwanted components, the mixture is thrown into

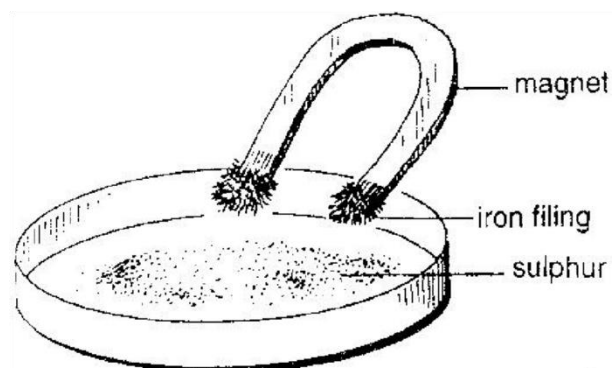
air where wind blows the husks away and the heavier grain fall back into the basket.

Decanting



- Decanting is a process used to separate immiscible liquids that have different densities. A distinct layer between the two constituencies is formed.
- The layer closer to the top of container – the less dense of two liquids is poured off, leaving the more dense liquid of the mixture behind. - Decanting can be used to
 - Separating a mixture of oil and water – oil floats on top of water and is carefully poured off
 - Separating a mixture of whey from milk – whey floats on top of

Magnetism



- Magnetism is used to separate a mixture of solids in which one of the components has magnetic properties.

Iron filings are attracted by the magnet, so they cling to the magnet, while the sulphur powder remains behind on the paper.

- Magnetism can be used in the following processes
 - Separation of metallic objects from grain before grinding – metal pieces can cause damage to the processing machinery. Magnets are used to attract any metallic objects before grinding.
 - Separation of metallic wastes for recycling – electromagnetic cranes are used to attract large loads of scrap iron and steel from wastes.

MATTER

STATES OF MATTER

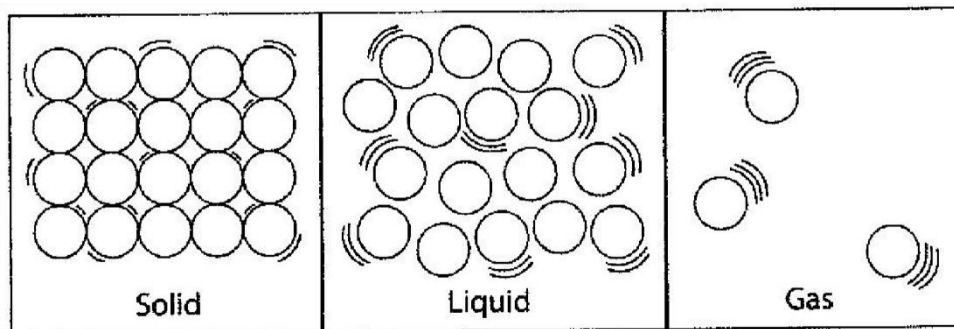
- Matter is anything that has mass and occupies space. It can be solid, liquid or gas.

Kinetic Theory

- The kinetic theory of matter states that matter is made of tiny particles that are always in constant motion.
- The behavior of these particles differs in the 3 phases

The three states of matter

- The properties of solids, liquids and gases can be explained by the kinetic theory.



Solid

- The particles in a solid are arranged in a fixed pattern or lattice. This accounts for the high density of solids.

- Strong intermolecular forces hold them together. This explains why solids have fixed volume and shape.
- The particles have very little kinetic energy and they only vibrate in fixed positions.
- Properties include definite shape, fixed volume, cannot flow, cannot be compressed and high density

Liquid

- The particles occur in clusters with the molecules slightly further apart compared to that of a solid. This accounts for the high density of liquids and the tendency of liquids to form droplets.
- The particles have some kinetic energy. The particles are free to move about between clusters but confined within the vessel containing it due to attractive forces between them. This explains why liquids have fixed volumes but take the shape of the vessels containing them.
- Properties include no fixed shape, less dense than solids, cannot be compressed, fixed volume and can flow.

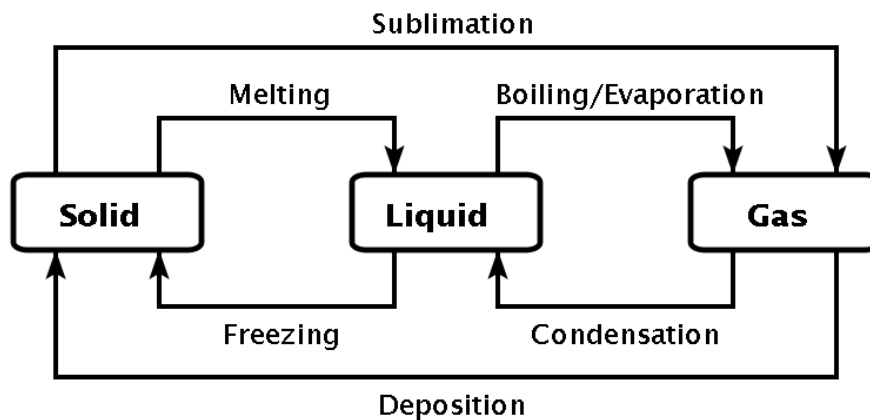
Gas

- The particles are very far apart in that the molecules will occupy any available space. This accounts for low density of gases.
 - The particles have lots of kinetic energy. The particles move at high speed, independent motion in random manner. Negligible attractive forces exist between them. This explains why gases have neither fixed volume nor shape.
- Properties include no fixed volume, shape and size, can flow and highly compressible.

Change of states

- If temperature is increased or reduced, matter changes its state. This is because a temperature change will affect the amount of kinetic energy.
- If a solid is heated, the energy will make the particles vibrate faster until the forces that hold them together are weakened and the particles move apart.
- If a gas cools, the particles lose energy and come closer together. Eventually strong forces develop between them.

- All changes of state involve an increase in kinetic energy or a decrease in kinetic energy caused by changes in temperature.



Melting

- When a solid is heated, its particles get more energy and vibrate more. This makes the solid expand. At melting point the particles vibrate so much that they break away from their position and solid turn to liquid.

Evaporation or boiling

- When a liquid is heated, its particles get more energy and move faster. They bump into each other more often, and bounce further apart. This makes the liquid expand. At the boiling point, the particles get enough energy to overcome the forces between them. They break away to form a gas.
- Some particles in a liquid have more energy than others. Even well below the boiling point, some have enough energy to escape and form a gas. This is called evaporation.

Condensation

- As a gas cools, its particles lose energy and move more slowly. The particles have less kinetic energy and move closer together. The gas changes state to form a liquid.

Freezing

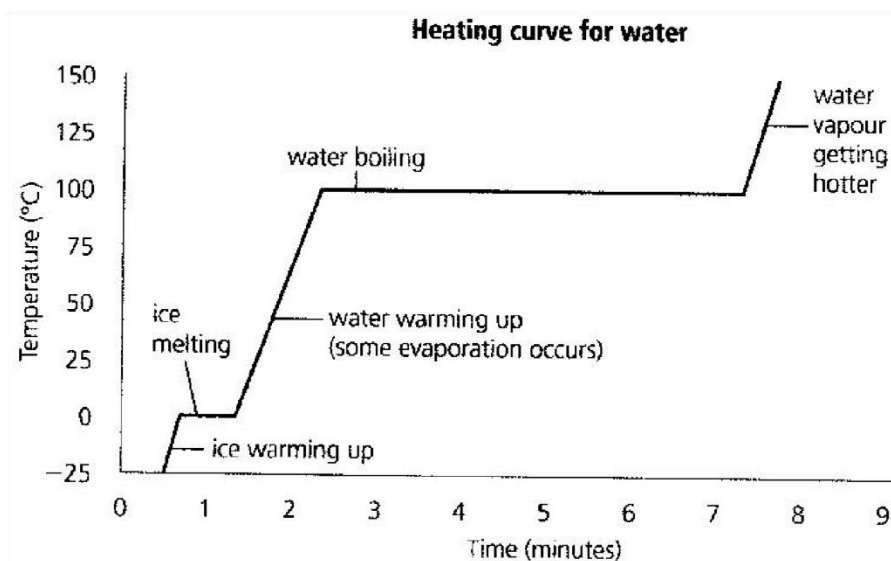
- When a liquid is cooled the particles have less kinetic energy. The particles in a liquid move slower and closer to each other. The liquid changes state to form a solid.

Sublimation and deposition

- With heating, some matter can change from solid state into the gas state, it is called sublimation. When cooled, the gas changes directly to a solid, this is called deposition.

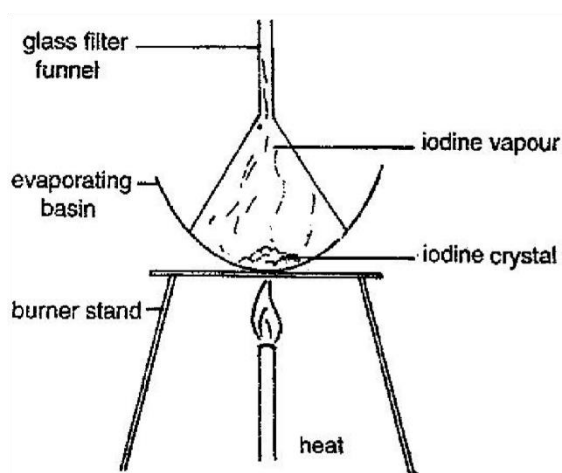
Experiment on heating water

- The temperature remains constant at 0°C as the ice melts. This is because heat supplied is being used to break bonds and change state from solid to liquid (latent heat of fusion).
- When all the ice has melted the temperature rises from zero to 100°C. Particles gain energy and move faster until they break away completely from each other.
- When boiling point is reached, energy being transferred is used to separate the liquid particles from each other.
- As heating continues, all the liquid is changed into a gas. While this is happening the temperature remains constant since heat absorbed is being used to break bonds in a liquid (latent heat of vaporisation)
- The temperature rises again after reaching boiling point. All the particles in the gas are now free to move and further heating raises temperature of the gas.



Experiment on heating iodine

- If iodine is heated in a crucible, it sublimes (sublimation). When the heat source is removed and the crucible is allowed to cool, the iodine cools and become deposited as a solid (deposition).
- Grey iodine changes directly to violet or purple vapor which collects on the cold part of the glass to form black shining crystals of iodine.
- Iodine vapor produced is toxic (poisonous) therefore the experiment should be done in a fume cupboard or close to a window.



SOLUBILITY

- The solubility of a *solute* is the maximum amounts of solute that can dissolve in a certain amount of *solvent* at a certain temperature until it become saturated.
- This means that only so much of a solute will dissolve and any more solute added after saturation will not dissolve even if the particles are further broken down or the solution is heated or stirred.
 - ✓ Solute is any substance that is dissolved in a liquid, to form a *solution*.
 - ✓ Solvent is a liquid that can dissolve other substances to form a solution.
 - ✓ Solution is a homogeneous mixture, which may be liquid, gas or solid, formed by dissolving one or more substances.

Factors that affect solubility

- The rate at which a solute will dissolve depends on particle size, temperature and how much you stir or shake the solution.

Temperature

- An increase in temperature of the solution, increases the solubility of the solute. A solute dissolve more quickly in hot water than it does in cold water because kinetic energy of particles increases as temperature increases thus increasing rate of solubility.

Stirring

- Particles dissolve more quickly when they are stirred because stirring increases the kinetic energy of particles causing them to collide more often with the solvent. Stirring therefore, allows the solute to dissolve faster.

Particle size

- Refers to how big the particles are e.g. fine and coarse salt. Small particles dissolve more quickly than larger ones because a larger surface area of the solute is in contact with the solvent. Many smaller particles will therefore dissolve faster than one large particle which has a smaller surface area in relation to its size. When the total surface area of the solute is increased (breaking a solute into small pieces), the solute dissolves more rapidly.

Elements, mixtures and compounds

Element

It is a substance made up of one type of atoms and cannot be split into two or more simpler substances by any chemical or physical means. Can exist as either atoms or molecules e.g. Hydrogen, Magnesium, Zinc, Iron, Copper.

Molecule

- It is the smallest electrically neutral particle of an element or compound that can exist on its own e.g. Hydrogen, Nitrogen, and Chlorine, Oxygen, Carbon dioxide and Water.

Mixture

- It is a substance which consists of two or more different substances physical intermingled e.g. salt water or iron filings and sulphur.

Compound

- It is a substance which consists of two or more elements chemically combined together and exists as molecules or ions e.g. copper sulphate, carbon dioxide, magnesium oxide.

Chemical reactions

- Chemical reaction takes place when heat is given out or taken in.
- The burning of magnesium ribbon in air is an example of chemical reaction where heat is taken in and once the reaction starts heat is given out.

Word equations

- Word equations are used to summarize chemical reactions. They show reactants used (left) and products formed (right).

Magnesium + Oxygen (reactants) → Magnesium Oxide (products) Iron + Sulphur →

Iron Sulphide

Physical and chemical changes

- A physical change is one in which the products have the same chemical properties as the reactants (no new substance is formed). Changes are mainly in physical properties or state of matter without accompanying change in composition e.g. evaporation, dissolving. A chemical change is one in which the products have chemical properties different from those of the reactants (new substance is formed). Substances are formed with entirely different properties and composition from the original materials e.g. combustion

Differences between physical and chemical changes

Physical Change	Chemical Change
No new substance is formed	New substance is formed
Properties do not change	New substance has different properties
Usually the change is easily reversible	Usually change is irreversible
Usually no energy is given out or absorbed	Usually heat energy is given out or absorbed
No change in mass	There is change in mass
A mixture is formed	A compound is formed

- When wax or ice is heated, no new substances are formed and their masses do not alter. The changes are physical. All changes of states are physical.
- If iron filings and sulphur powder, the iron in the mixture is uncombined with the sulphur and will be attracted to a magnet. But if the mixture is heated, the iron and the sulphur which is not attracted by a magnet. Iron sulphide is a compound which can only be separated by chemical means.
- When Magnesium or mealie meal is heated, a new substance is formed. In many chemical changes, energy is given out as heat, sound or light. It is difficult to change the new substances back to the original Magnesium.

THE PERIODIC TABLE

Classification of elements

- There are 118 elements on the periodic table
- The periodic table is an arrangement of elements in order of their atomic numbers.
- The periodic table shows the names, chemical symbols and atomic numbers for each element.

The elements are organised into columns (groups) and rows (periods) based on the structure of their atoms and their properties

- The elements in the periodic table are arranged into 3 main categories; metals, metalloids and non-metals.

Groups

- There are 8 main groups of elements i.e. I, II, III, IV, V, VI, VII and O. elements between group II and III i.e. group 3-12 are called transition elements.
- Elements in the same group typically have the same number of electrons in their outermost electron shell e.g. elements with one electron in their outermost shell are in group 1.
- Since electrons in the outermost shell are usually the ones that take part in a chemical reaction, elements in the same group undergo the same chemical reactions. - Some groups have special names;
 - ✓ Group I – alkali metals
 - ✓ Group II – alkaline earth metals
 - ✓ Group VII – halogens
 - ✓ Group VIII – noble gases

Periods

- Periods are numbered 1 to 7
- Electrons in an atom are placed in electronic shells and each shell can contain a separate number of electrons.
- Periods show the number of shells the elements have. For all elements in the same period, electrons are being added to the same shell as we move across the period from left to right.

Metals and non-metals

- Metals are mostly found on the left and in the middle of the periodic table.
- The elements around the dividing lines are referred to as metalloids and have properties that are similar to both metals and non-metals.
- Non-metals are found on the right of the periodic table
The zigzag line separates metal from non-metals, with the non-metals on the right of the line, except for Hydrogen. So there is a change from metal to non-metal, as you go across a period.

Common elements

- Each element has its own chemical symbol and below is a list of the first 20 elements of the periodic table based on their atomic number and their symbols.
- Each element has its own name, symbol, atomic number and its own position on the periodic table

Group	I	II	III	IV	V	VI	VII	VIII
Period								
1	${}_1\text{H}$							${}_2\text{He}$
2	${}_3\text{Li}$	${}_4\text{Be}$	${}_5\text{B}$	${}_6\text{C}$	${}_7\text{N}$	${}_8\text{O}$	${}_9\text{F}$	${}_{10}\text{Ne}$
3	${}_{11}\text{Na}$	${}_{12}\text{Mg}$	${}_{13}\text{Al}$	${}_{14}\text{Si}$	${}_{15}\text{P}$	${}_{16}\text{S}$	${}_{17}\text{Cl}$	${}_{18}\text{Ar}$
4	${}_{19}\text{K}$	${}_{20}\text{Ca}$						

CONCENTRATIONS

- Concentration is the amount of solute that is dissolved in a solvent.
- If the identity of the solute and solvents in a solution are known, the concentration of the solution can be determined.

Experiment to determine concentration by colour intensities

- Put KMnO_4 crystals in test tube A and add 20cm^3 of water. Stir the solution until all crystals have dissolved.
- Use a pipette to measure 2cm^3 of the solution in test tube A and transfer it to test tube B. Add 18cm^3 of water.
- Repeat for test tube C and D, each time taking 2cm^3 of solution from the previous test tube.
- The colour intensities of the various solutions will range from dark or deep purple colour to a light or very light pink colour because a 1:10 dilution was created each time since

each successive concentration is $\frac{1}{10}$ of the previous concentration. The final dilution appears much lighter in colour than the others.

- To determine the % concentration of a solution, the following equation is used;

$$\% \text{ composition} = \frac{\text{total volume of solution}}{\text{volume of solvent}} \times 100$$

- Higher concentration of a substance will give solutions of darker or deeper colours. As the solutions are diluted, their intensities decrease.

ACIDS, BASES AND SALTS

ACIDS

- An acid is a compound which when dissolved in water, forms hydrogen (H^+) ions as the only positively charged ions.
- Strong acids are completely ionized e.g. sulphuric acid, Hydrochloric acid and Nitric acid. Weak acids are only slightly ionized in dilute solutions.
- Examples of acids hydrochloric acids (HCl), sulphuric acids (H_2SO_4) and nitric acid (HNO_3)

Properties of Acids

- Turn litmus paper red, methyl orange red, phenolphthalein colorless and universal indicator red or orange
- Have a pH below 7
- They are corrosive
- Have a sour taste

BASES

- A base is a compound which contains oxide or hydroxide ions.
- An alkali is a compound which when dissolved in water forms hydroxide (OH^-) ions. They dissolve in water to form alkaline solutions,
- Examples of bases all metal oxides, all metal hydroxides and ammonia

Properties of Bases

- Turn litmus paper blue, universal indicator blue or purple, phenolphthalein pink and methyl orange yellow.
- Taste bitter

- Have a slippery feel
- Strong bases are caustic e.g. potassium hydroxide

Identifying acids and bases using litmus paper

- Litmus is a purple dye that can be in form of a solution or in the form of red or blue paper - Litmus is called an indicator because it indicates whether a substance is a base or an acid.
- Substances that are neutral are neither acidic nor basic.
- Both red and blue litmus paper should be used to test a solution. If the colour does not change, it means that it does not indicate that the substance is an acid or base
- Dip litmus paper in five test tubes with each of the following substances i.e. sodium hydroxide, dilute hydrochloric acid and distilled water, tap water, copper sulphate
- Record the results in the table below and draw conclusions about the colour change in each substance.

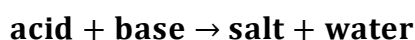
Substance	Effect on red litmus paper	Effect on blue litmus paper	Acid or alkali
Hydrochloric acid (HCl)			
Sodium hydroxide (NaOH)			
Tap water			
Distilled water			
Copper sulphate solution			

Acid – Base reactions

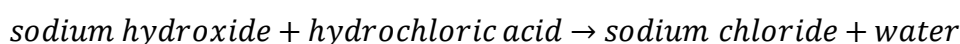
- When an acid react with a base, it is a neutralization reaction.
- During the reaction, salt is always produced. Type of salts produced depends on the acid used i.e.

Acid	Salt formed
Hydrochloric acid (HCl)	Chloride
Sulphuric acid (H ₂ SO ₄)	Sulphate
Nitric acid (HNO ₃)	Nitrate

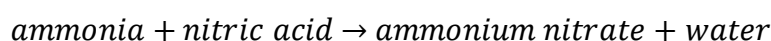
- Bases release hydroxide ions (OH⁻) while acids release (H⁺). During neutralization H⁺ ions combine with OH⁻ ions to form water



- A reaction between sodium hydroxide and hydrochloric acid gives sodium chloride and water



- A reaction between ammonia and nitric acid give ammonium nitrate and water



- A reaction between copper oxide and sulphuric acid gives copper sulphate and water *copper oxide* + *sulphuric acid* → *copper sulphate* + *water*

INDUSTRIAL PROCESSES

PRODUCTION OF PEANUT BUTTER

Shelling

- Consists of removing the peanut shell with the least damage to the seeds inside.
- The peanuts are passed between a series of rollers where the peanut shells are gently cracked. The cracked peanuts are then repeatedly passed over screens and blowers, where they are shaken, gently tumbled and air-blown until all the shells are removed.
- Another option would be to shell the peanuts by hand and using a winnowing basket to separate the lighter shells from the peanuts

Roasting

- Peanuts are roasted in special ovens at 180°C for about 10 minutes. This process destroys certain enzymes in the peanuts that may produce bad flavors. Roasting also enhances the color, flavor and texture of the peanuts.
- Roasting at home is done using a pan over a wood fire or stove.

Grinding

- The peanuts are lightly rubbed between rubber belts to remove their outer skins around each peanut. The result is peanuts that are paler in colour than they were with skins. This is called blanching.
- The splitted peanuts are then ground in a grinding machine twice. First time reduces the nuts to a medium grind and the second time produce a fine smooth texture.
- The peanuts are ground alone first and then ingredients are added to them like salt, sweetener and stabilizer. The stabilizer stops the oil from separating out of the butter
- In rural areas splitted peanuts are pounded in a mortar (duri) using a pestle (mutswi) followed by grinding on the grinding stone (guyo).

Packaging

- The peanut butter is cooled and taken to filling machine where the correct amount of peanut butter is poured into empty jars.
- An automatic capping machine places a lid on each jar. The jars are labeled to show the contents mass and nutritional values. The jars are sealed to prevent it from being exposed to the air. They are packed ready to be transported to stores for sale.

Production of oil from peanut butter

- Pour the peanut butter in a bowl and cover the lid tightly and put in a refrigerator for at least a day. The oil will rise on top of the butter.
- Strain the oil into a separate bowl. When the oil is completely clean after several times of straining, pour it in a clean bottle.

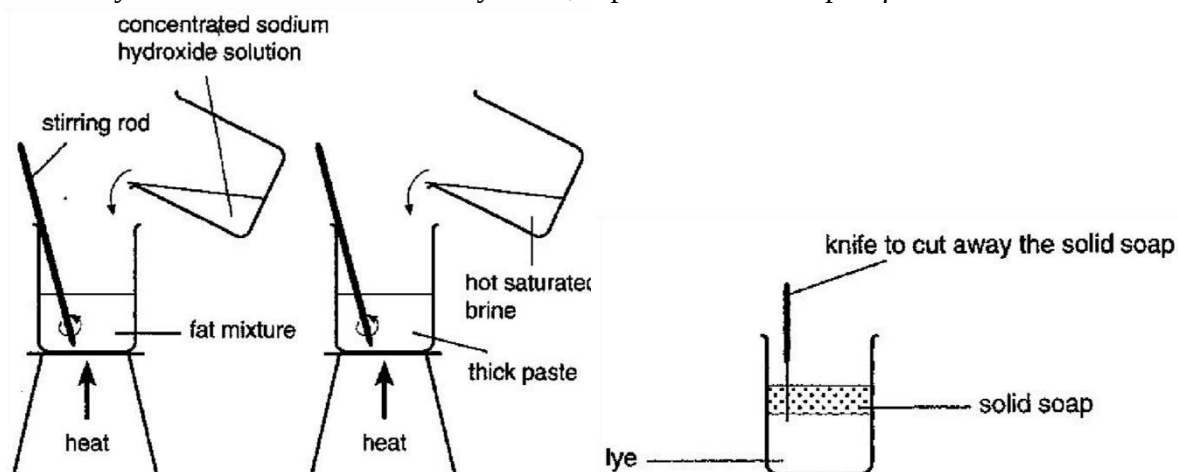
Uses of oil

- Cooking oil – it has a higher burning and smoking point
- Baby care products such as nappy rash cream
- Baking
- In salad making
- Skin moisturizer
- Massage oil – it has a light and nutty smell
- Healthy oil because it can be produced at low temperature.

PRODUCTION OF SOAP

Saponification

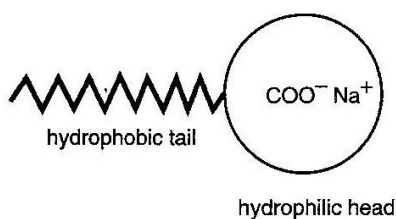
- Soaps are made by the action of alkalis on fatty acids, a process called *saponification*.



- Oils and fats are heated by steam with sodium hydroxide. The fats are broken down, leaving glycerol and the sodium salt of the long chained acid.
- The sodium salt of the acid can be used as soap.
- Sodium chloride is added to separate the soap as an upper layer (the salting out of soap).



- The soap molecule has a hydrophilic head and the hydrocarbon tail is hydrophobic. When washing with soap and water, the hydrophilic end is attracted to the water molecules while the hydrophobic end mix with the oils from the dirty object, detaching them from the object and attach themselves to the soap



Stages in the production of soap

Boiling

- The alkali and fat are boiled in a large kettle (steel tank). After boiling the mass thickens as the fat reacts with the alkali producing soap and glycerine

Separation or Salting

- Salt is added to the mixture to cause the soap product to rise to the top of the kettle and allow the glycerol to settle at the bottom. In this way the glycerol is simply tapped off from the bottom of the kettle

Refinement

- To remove the small amounts of fat that have not saponified, a strong caustic alkali solution is added to the kettle. The mass is brought to the boil again and react the last of the fat turns to soap

Pitching

- Involves boiling the soap again with added water. The mass eventually separates into two layers. The bottom layer called *nigre* containing most of the water, impurities that such as dirt and salt. The top layer, called the *neat soap* is a liquid layer that contains all the purified soap (70%) and some water (30%). The neat soap is poured into moulds and allowed to cool.

OXIDATION AND REDUCTION

Oxidation

- Oxidation is gain of oxygen
 - Sulphur is oxidized to sulphur dioxide by gaining oxygen.
$$\text{sulphur} + \text{oxygen} \rightarrow \text{sulphur dioxide}$$

Reduction

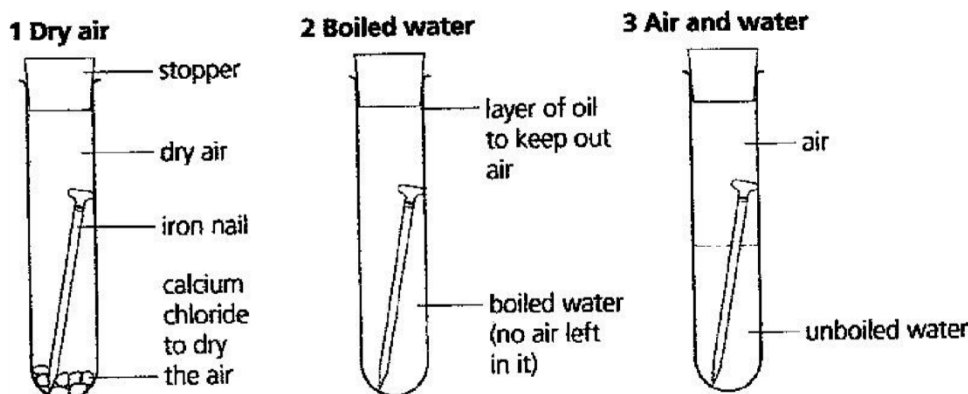
- Reduction is the removal of oxygen
 - Iron oxide is reduced to iron by losing oxygen to carbon
$$\text{iron oxide} + \text{carbon} \rightarrow \text{iron} + \text{carbon dioxide}$$

Rusting

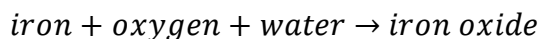
- Rusting is the corrosion of iron i.e. the formation of iron oxide.
- When iron comes into contact with oxygen, a chemical reaction takes place, producing iron oxide (rust)

- When the rust forms, the flakes of rust fall off the surface of the metal, exposing more iron to the factors that cause rusting, resulting in further rusting of the object. In this way, the iron is gradually corroded hence deteriorates and destroyed over time by rust. **Conditions necessary for rusting**
- Water
- Oxygen

Experiment on conditions needed for rusting



- Stand three identical nails in three test tubes. - Prepare the test-tubes as below so that
 - Test tube 1 - Contains dry air
 - Test tube 2 - Contains water but no air
 - Test tube 3 - Has both air and water
- Leave the test-tubes for several days
- After several days the nails in test 1 and 2 shows no signs of rusting. The nail in test tube 3 has rust on it. This is because rusting requires water and oxygen. In fact the iron is oxidized in this reaction.
- The anhydrous calcium chloride in test tube 1 absorbed moisture thus drying the air.
- The water in test tube 2 was boiled to remove air from it. A layer of oil prevents the entry of air into the tube.



Methods of preventing rusting

1. Surface protection – covers the metal with a layer of substance e.g.
 - Painting or oiling – involves applying a thin coat of paint which will not allow both air and water from coming into contact with the metal. The paint must be kept in good condition because any scratch in the paint surface will expose the iron to air and moisture causing it to rust.

2. Sacrificial protection – covering metal with thin layer of another more reactive metal e.g.

□ Electroplating – iron is often coated with other metals to prevent it from rusting
e.g. plating with copper, chromium and nickel.

□ Galvanizing – iron or steel is coated with zinc by dipping the sheet of iron in molten zinc. Zinc corrodes before iron but even when corroded it forms a layer of zinc oxide which prevents the iron from being corroded.

ORGANIC CHEMISTRY

FUELS

- A fuel is a material that can be burnt to give out heat energy or have chemical energy.

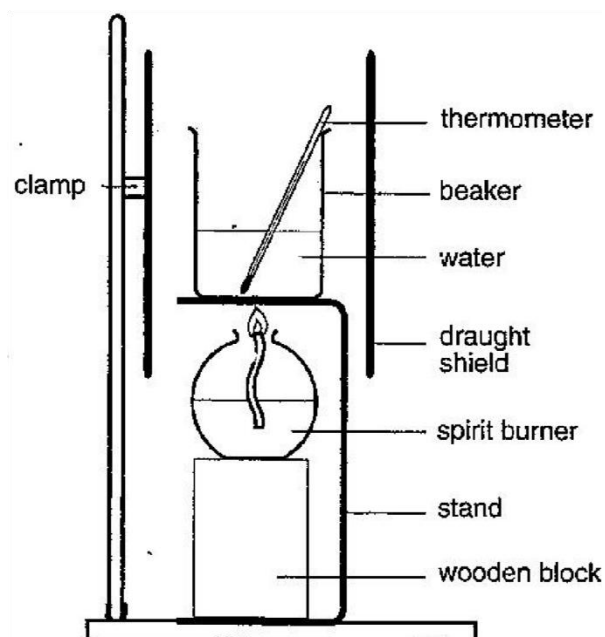
Types of fuels

- Fuels are found in three forms i.e.
 - **Solid fuels** e.g. wood, charcoal, coke and coal. They are relatively cheap and mostly readily available. They are easy to store. Have low heating efficiency. Extraction can cause environmental degradation and have high carbon emissions.
 - **Liquid fuels** e.g. petrol, diesel, paraffin and ethanol. More expensive than solids. Are less available and can be stored more compactly and in a closed container. High heating efficiency and are more flammable than solid fuel. Release carbon emissions when burnt.
 - **Gaseous fuels** e.g. methane, ethane, hydrogen and coal gas. Costly and less available. Needs to be stored in leak proof container. High heating efficiency and have very risk, incredibly flammable and leaks can go undetected. Burns without smoke but fracking for natural gas causes land and water pollution.

Thermal efficiency

- Thermal efficiency is the measure of the heat content of a fuel.
- Carbon in fuels burns to produce heat energy.
- The efficiency of a fuel can be measured by the rate at which the fuel heats up a substance to certain temperature.

Comparing the efficiency of different fuels



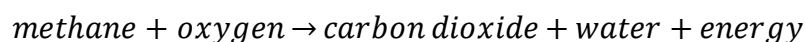
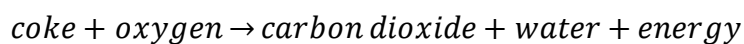
- Equal volumes of liquid fuels were used to heat equal volumes of water for the same period of time.
- Temperature readings were taken at the beginning and at the end of the heating process.
- The temperature of water heated using methylated spirits increased less than that of paraffin.
- Paraffin has a higher thermal efficiency than methylated spirit. This is because paraffin has a higher carbon content which is evident because the flame has more soot.
- The more the carbon content the fuel contains, the higher the heating efficiency e.g. coke and charcoal have higher carbon content than wood and coal and therefore give out more heat energy when burnt.

Type of fuel	Examples	Heating value
Solid	Wood	17
	Coal	25
	Coke	28
	Charcoal	33
Liquid	Ethanol	30
	Petrol	45
	Paraffin	48
	Diesel	55

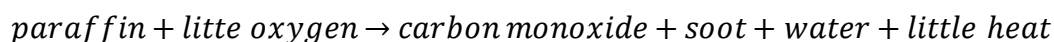
Gas	Coal gas	43
	Biogas	40
	Butane	50
	Methane	55
	hydrogen	60

Complete and Incomplete combustion

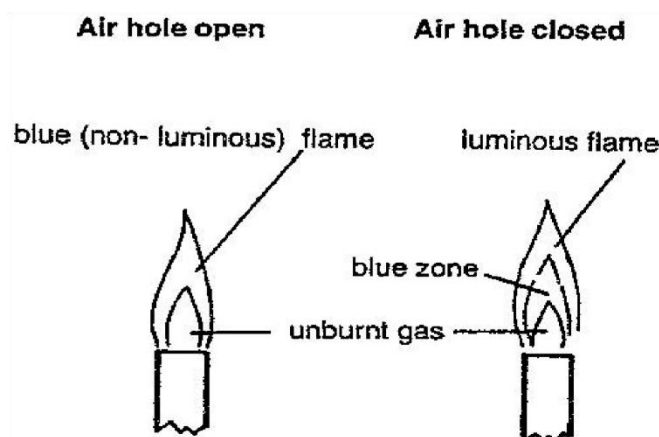
- Combustion means the burning of a fuel to give energy.
- Many fuels contain carbon and hydrogen. When the fuels burn the carbon is oxidized to carbon dioxide while hydrogen is oxidized to water.
 - Complete combustion occurs when a fuel burns in a plentiful supply of oxygen. All carbon is burnt.



- Incomplete combustion occurs when the amount of carbon in a fuel exceeds the amount of oxygen available. Not all carbon is burnt is given off as soot while some has partial oxygen and is turned into carbon monoxide. Carbon monoxide is a poisonous gas which can kill if excessively inhaled. Incomplete combustion causes a lot of air pollution because of the soot and carbon monoxide.



Demonstrating complete and incomplete combustion using burners



- The blue zone on the Bunsen burner indicates complete combustion and the yellow flame indicates incomplete combustion.
- The air hole on the burner controls the amount of oxygen that mixes with the gas.
- When air hole is open, air enters the tube and mixes with the gas which therefore burns quickly and completely. The flame is small and has a dominant non-luminous blue flame.
The flame is clean and very hot and can heat water very rapidly.
- When the air hole is closed, no air enters the tube and the gas mixes with little air hence incomplete combustion. The flame is larger and has a dominant yellow luminous flame which is sooty which blackens the underside of the beaker. The flame is not very hot and gives little energy such that it takes longer to raise temperature of water

Effects of complete and Incomplete Combustion

1. Deforestation

- Uncontrolled cutting down of trees for firewood causes deforestation which will result in soil erosion and loss of top soil causing river siltation.
- Deforestation upsets the balance of nature i.e. plant and animal habitats are destroyed so that they cannot support the normal animal population.
- Deforestation causes an increase in the carbon dioxide level in the atmosphere because there is now less vegetation to remove atmospheric carbon dioxide during photosynthesis.

2. Pollution

- Incomplete combustion of fuels like petrol in motor engines leads to emission of soot and carbon monoxide. Carbon monoxide is a poisonous gas while soot causes smog over cities.
- Fuels also give off volatile substances which pollute the air most of which may cause cancer.
- The nitrogen and sulphur from fossil fuels are oxidised into acidic oxides that make rain water more acidic than it should be. Plants are damaged by acid rain, fish in lakes and rivers cannot tolerate acidic conditions hence are destroyed.

3. Global Warming

- Large amounts of carbon dioxide build up and a layer of gas is created in the atmosphere which stops heat from escaping the atmosphere and is reflected back to earth resulting in global warming. Rise in earth's temperature will lead to fertile land becoming dry and uninhabitable. In arctic regions ice caps slowly melts resulting in flooding.

PHYSICS

COMBINED SCIENCE NOTES



FORM 1 & 2
PHYSICS

TOPICS

- ❖ **DATA PRESENTATION**
- ❖ **MEASUREMENTS**
- ❖ **FORCE**
- ❖ **MECHANICAL SYSTEMS**
- ❖ **ENERGY**
- ❖ **MAGNETISM**
- ❖ **ELECTRICITY**

DATA PRESENTATION

WAYS OF PRESENTING DATA

- Data is any information such as facts and statistics gathered by scientist during experiment or during a research.
- Scientists interpret data (i.e. understand and explain the meaning of) and it to make conclusions about their experiment.
- Data gathered need to be presented in a certain way which is not time consuming e.g. visual or graphic.
- Data can be qualitative data (observations) or quantitative data (statistical data).

Tallies

- Tally marks are a unary numeral system. They are a form of numeral used for counting. They are most useful in counting or tallying ongoing results, such as the score in a game or sport, as no intermediate results need to be erased or discarded.
- However, because of the length of large numbers, tallies are not commonly used for static text.
- Tally marks are typically clustered in groups of five for legibility. The cluster size 5 has the advantages of easy conversion into decimal for higher arithmetic operations and avoiding error, as humans can far more easily correctly identify a cluster of 5 than one of 10.

- Tallies are done as follows 

A tally of mass of form one learners

Mass (kg)	Number of learners	tally
31 – 35	1	I
36 – 40	6	IIII I
41 – 45	11	IIII IIII I
46 – 50	8	IIII IIII
51 - 55	4	IIII

Tables

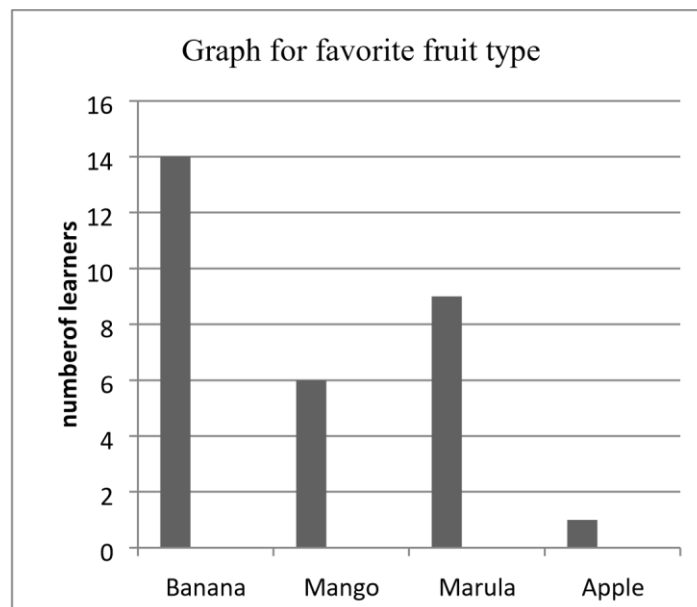
- Presenting data in a table helps to make it clear and easy to read and understand.
- Tables should always have a descriptive heading
- Each column and row should be labelled and can have a unit of measurement representing all data on the column if applicable. For example;

Favourite fruit types for form one learners

Types of fruit	Number of learners
Banana	14
Mango	6
Marula	9
Apple	1

Bar graphs

- A bar graph is a visual display of data on a graph. Bar graphs have vertical bars of different heights on a pair of axes.
- Bar graphs are used when the data is in groups or categories e.g. days of the week, types of transport, type of fruits e.t.c.
- We can use the information presented in the table above to draw a bar graph



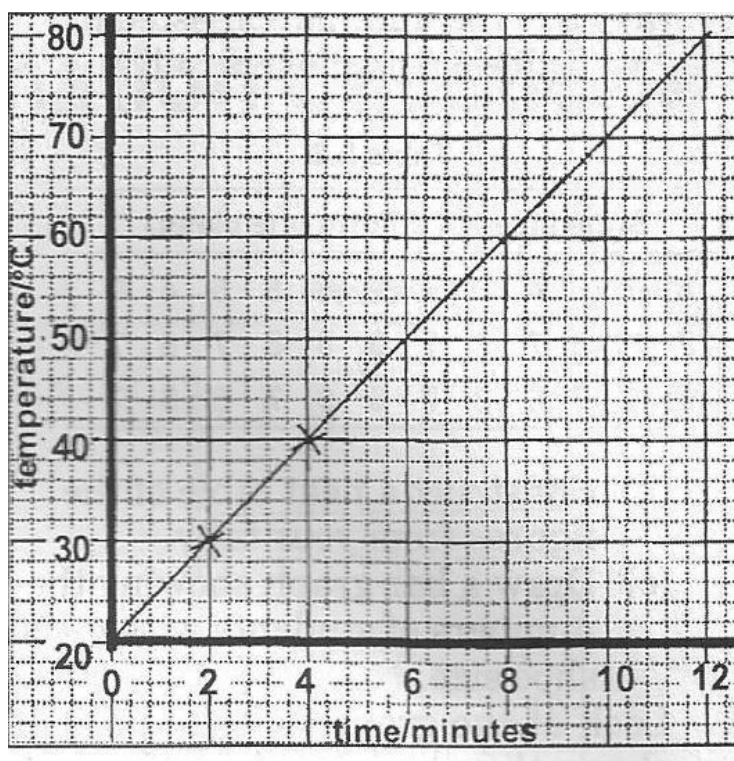
- The heights of the bars shows the values represented in the table

- From the graph we can see that bananas are the most popular fruit and apples are the least popular.

STRAIGHT LINE GRAPHS

- Shows a variable that change with time e.g. temperature change as water is heated,
- It has x-axis and y-axis
- Points to note when drawing a line graph include;
 - Draw the lines to show the x-axis and y-axis
 - Label the axis
 - Select suitable scale for both the x-axis and y-axis
 - Mark points accurately using the following mark • or +
 - Draw a smooth line to join the points i.e. draw a line of best fit using a sharp pencil

A line graph showing temperature changes with time



- With such a straight line we can conclude that the temperature is directly proportional to the time. As time increases temperature also increase
- The graph can be used to find the temperature at 3 and 7 minutes

Examples 2 A student carried out an investigation the effect of height on pressure of a liquid.

Table below shows results obtained from the experiment.

depth/cm	10	20	30	40	50
height/cm	0.5	1.0	1.5	1.8	2.5

- Plot a graph of height against the depth
- Find , using the graph, the height at depth of 24cm
- Describe, from the graph, the relationship between height and depth

Example 3 An experiment was done to investigate the relationship between the distances travelled over the time taken for a car to move along a track. The following data was collected.

Time/s	0	2	4	6	8	10
Distance/m	0	5	10	15	20	25

- (a) Plot a graph of distance travelled against time taken. (b) What is the relationship between the variables

MEASUREMENT

PHYSICAL QUANTITIES

Estimating Quantities

- An estimate is a guess very close to actual based on knowledge or rough calculations and it can be done before actual measurement. Different people produce different estimates for a given quantity.
- Estimate the length of the following and record your estimations in metres
 - Width of the classroom door
 - The length of the classroom walls
 - Height of your desk
- Estimate and record estimates in kg / g of mass of the following
 - A science textbook
 - A pen
 - A beaker
- Estimate and record your estimations in °C of temperature of
 - Cold water

(b) Warm water

(c) Hot water

- Estimate and record your estimation in seconds or minutes of time it takes to

(a) Walk the length of the class

(b) Take your book out of your satchel and place it on your desk

(c) To boil 100ml of water

- Record all observations in the table below

Item	Estimation	Actual Measurement	Accurate /not accurate

ERRORS IN MEASUREMENT

- Errors occur in all physical measurements and there are two common errors that could occur when taking measurements i.e.

Parallax error

- It is an error in reading an instrument due to the incorrect position of the eye.
- To avoid parallax error, the person taking measurement must make sure that their line of sight is directly in line with the instrument's pointer and scale.

Zero error

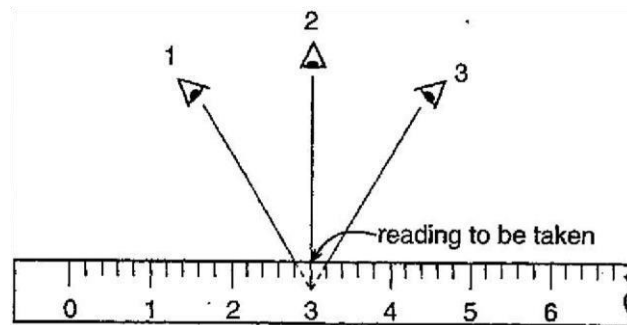
- Is caused by incorrect positioning of the zero point.
- The pointer on the instrument must be exactly positioned near to zero on the scale

PHYSICAL QUANTITIES AND SI UNITS

- Physical quantity is a property of an object or other substance that can be measured using an appropriate measuring instrument.
- Each type of measurement is done in a special unit, set as a standard to be used by scientists and other people so that they communicate effectively amongst themselves.
- SI units stand for **international system of units**.
- The International System (SI) units of measurements are used to measure physical quantities.

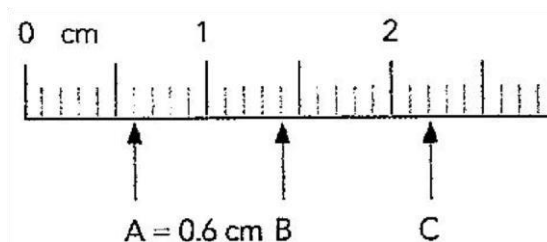
Reading an instrument scale

- It is important to be accurate when taking measurements.
- Errors in measurement can be minimised to ensure accurate measurement
- The eye must be correctly positioned in order to take correct readings
- In diagram below position 2 will give correct reading. Position 1 will give a reading that is less than the actual measurement and position 3 will give a reading that is greater than the actual measurement.



- To read the scale properly, count the number of divisions between zero and one. For example, in the scale below, there are ten divisions between zero and one. Divide 1 by the number of subdivisions to find out what each subdivision measures, in this

case, each subdivision represents 0.1cm.



- Therefore reading at position A is 0.6cm; at position B is 1.4cm and position C is 2.2cm

Units including SI Units

Physical quantity	SI unit	Instruments
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Length	Metre (m)	Metre rule, measuring tape
Mass	Kilogram (kg)	Balance
Time	Second (s)	Clock, stop watch
Temperature	Kelvin (K)	Thermometer

CONVERTING UNITS

Prefixes of SI units

- Kilo- (k) 1000
- Milli- (m) 0.001
- Centi- (c) 0.01

Length

- The SI unit is the metre and other units include centimetre (cm) and millimetres (mm).

$$100\text{cm} = 1\text{m}$$

$$10\text{mm} = 1\text{cm}$$

- When converting;
 - (a) Metre to centimetre multiply metres by 100
 - (b) Centimetre to metres divide by 100
 - (c) Millimetre to centimetre divide by 10
 - (d) Centimetre to millimetre multiply by 10

Mass

- The SI unit is the kilogram and other units include the gram and the milligram

$$1\text{kg} = 1000\text{g}$$

$$1\text{g} = 0.001\text{kg}$$

- When converting;
 - (a) Kilograms to grams multiply by 1000
 - (b) Grams to kilograms divide by 1000

Time

- SI unit is the second. The other units include minutes and hours

$$1\text{h} = 60\text{mins}$$

$$1\text{min} = 60\text{s}$$

$$1\text{h} = 3600\text{s}$$

- When converting;
 - (a) Hour to minutes multiply by 60
 - (b) Minutes to hours divide by 60
 - (c) Seconds to minutes divide by 60
 - (d) Minute to seconds multiply by 60

Temperature

- SI unit is the Kelvin. The other units include degrees Celsius ($^{\circ}\text{C}$) and degrees Fahrenheit ($^{\circ}\text{F}$)

$$0^{\circ}\text{C} = 273\text{K}$$

- When converting;
 - (a) Degrees Celsius to Kelvin add 273
 - (b) Kelvin to degrees Celsius subtract 273

MEASURING PHYSICAL QUANTITIES

Measuring mass of liquid

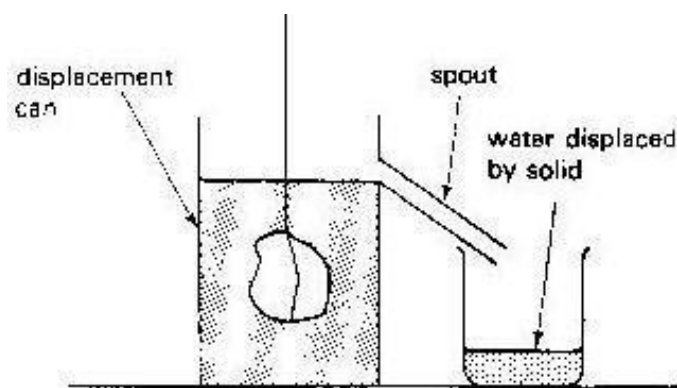
- A liquid cannot be weighed directly without placing it in a container.
- The mass of an empty beaker is found on a balance.
- A known volume of the liquid is transferred from a measuring cylinder into the beaker. The mass of the beaker plus liquid is found.
- Therefore the mass of liquid is obtained by subtraction as follows.

$$\text{mass of water} = \text{mass of beaker and water} - \text{mass of empty container}$$

Volume of irregular objects

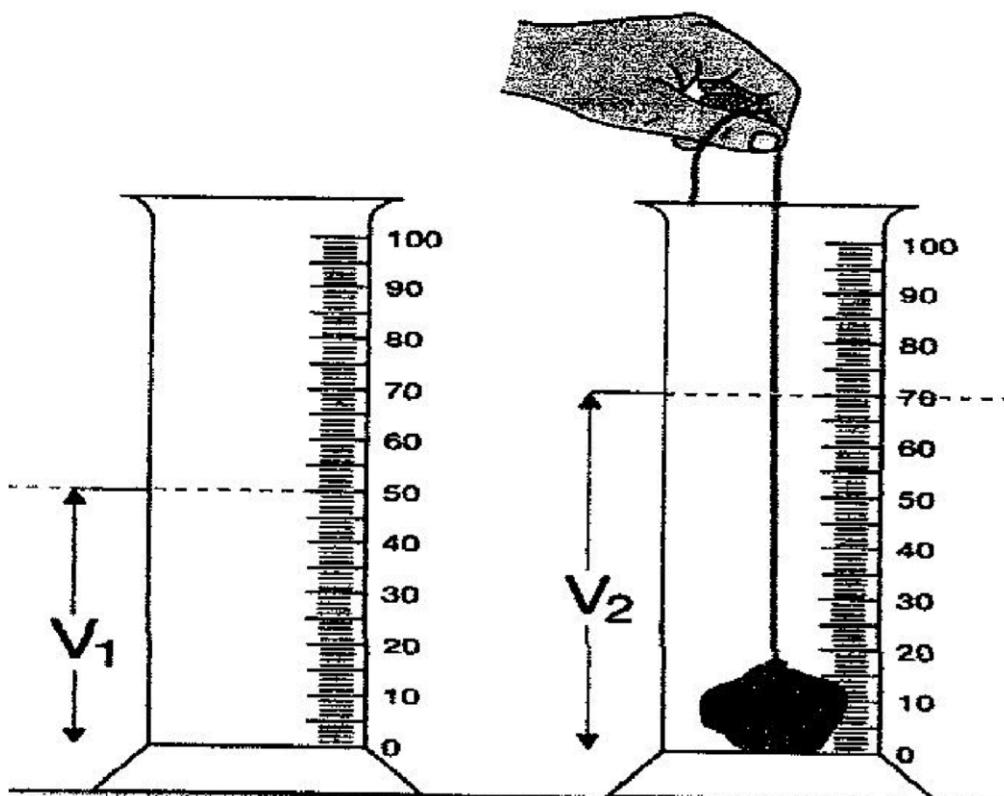
- Volume of an irregular object can be found by displacement method.
- When a solid is immersed in water, it displaces its own volume of liquid in which it is immersed.

- Fill the overflow can so that water is level with the bottom of the spout. Then place the object in the can, collecting the water which overflows.
- Measure its volume; this equals the volume of the object.



- If the displacement can is unavailable, use a measuring cylinder instead. Fill it about half-full with water; read and record the volume of water. (initial volume)
- Put the object in water so that it is completely covered; read and record the total volume of water and object (final volume) - Calculate the volume of the object

$$\text{volume of object} = \text{final volume} - \text{initial volume}$$



Original volume of water = 50cm^3

Final volume of water = 70cm^3

$$\text{Volume of stone} = 70\text{cm}^3 - 50\text{cm}^3 = 20\text{cm}^3$$

Mass of small object

- If the object is very small, it is very difficult to measure its mass e.g. the mass of one bean seed or drawing pin.
- Measure the mass of a large number of the objects and then divide the mass by number of objects to get mass of one object

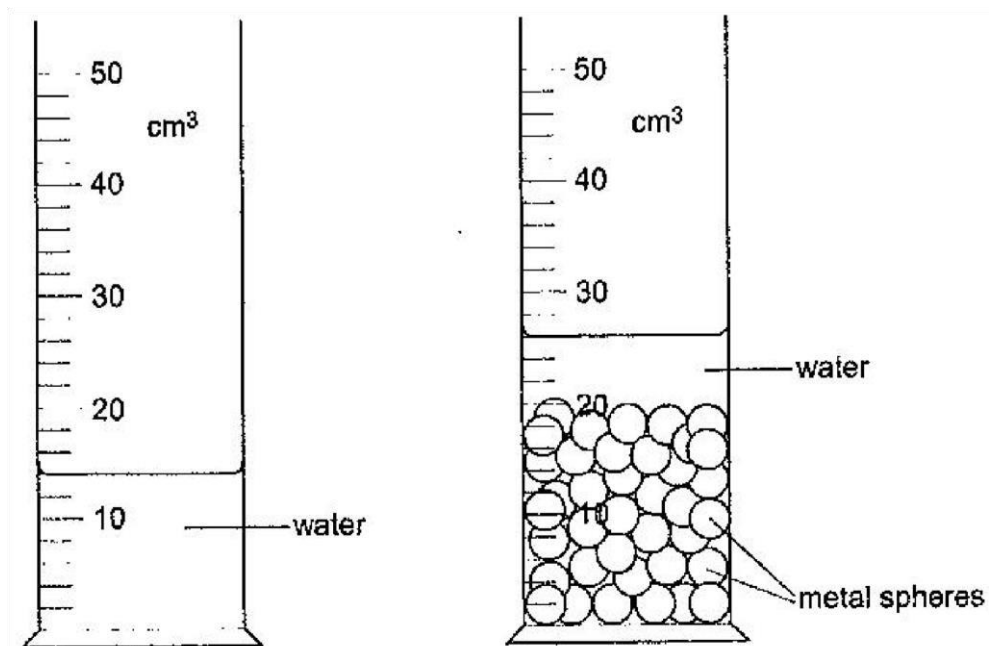
$$\text{mass of one seed} = \frac{\text{mass of seeds}}{\text{number of seeds}}$$

Volume of small objects

- Fill a measuring cylinder with water and record the volume.
- Place a large number of objects e.g. 50 seeds into the water in the measuring cylinder.

Record the new volume.

- Calculate the volume of the large number of the objects by subtracting the volume of water from that of water and seeds.
- Calculate the volume of one object by dividing the volume by number of objects



$$\text{Volume of water} = 14\text{cm}^3$$

$$\text{Volume of water} + 50\text{seeds} = 26\text{cm}^3$$

$$\text{Volume of 50seeds} = 26 - 14 = 12\text{cm}^3$$

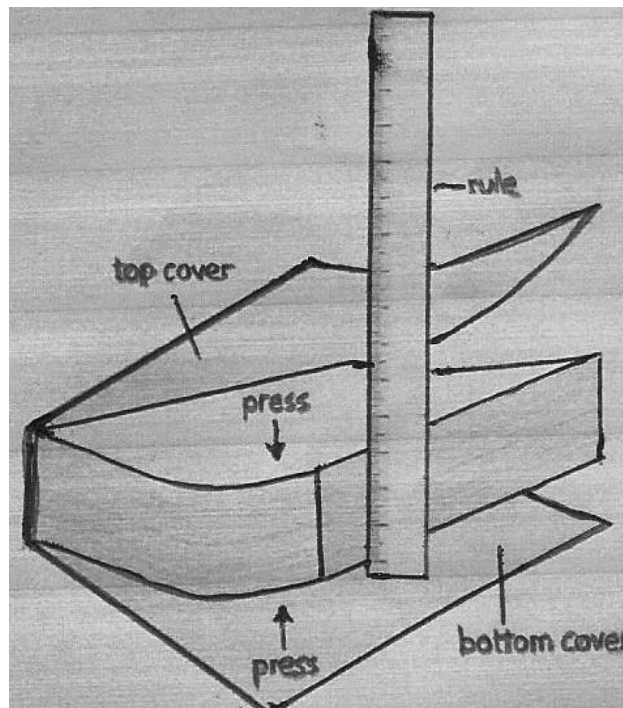
$$\text{Volume of 1 seed} = 12/50 = 2\text{cm}^3$$

Thickness of small objects

- Separate the cover of the book from the rest of the pages
- Press together the pages and then count the number of sheets in your book.
- Measure the thickness of the sheets.
- Divide the thickness of the sheet by number of sheets to get thickness of one sheet.

$$\text{thickness of one sheet} = \frac{\text{thickness of sheets}}{\text{number of sheets}}$$

- Since the thickness of one sheet is so small you can convert cm to mm



Example 1 A pile of exercise books without covers is 20cm high. There are 30 exercise books each with 46 pages. What is the thickness of one sheet of paper?

$$\text{Number of pages} = 30 \times 46 = 1380$$

$$\text{Number of sheets} = \frac{1380}{2} = 690$$

$$= \frac{20\text{cm} \times 10\text{mm}}{690} = \frac{200}{690}$$

$$\text{Thickness of one sheet} = 0,29\text{mm}$$

DENSITY

- Density is mass per unit volume of a substance

$$\text{Density (D)} = \frac{\text{mass (m)}}{\text{volume (V)}}$$

- Units can be g/cm³ or kg/m

Example 1 Calculate the density of glass if 120cm³ of glass has a mass of 300g

Example 2 A cylinder of aluminium has a radius of 7cm and a height of 20cm. The mass of the cylinder is 8.316kg. Calculate the density of aluminium

Example 3 A beaker has a mass of 48g. When 120cm³ of copper sulphate solution are poured into the beaker it is found to have a mass of 174g. Calculate the density of the copper sulphate

FORCE

EFFECTS OF FORCES

- A force is a push or a pull; a squeeze or a twist - Effects of forces include;

- Distortion or deformation (change in shape and size) – a force can change the shape of a solid object e.g. when you squeeze a cool drink can, the force exerted on the can by the hand causes the can to change shape.
- Change in speed – a force can cause the speed of a moving object to change. It can cause the object to accelerate and move faster or decelerate and moves slower e.g. when riding a bicycle the harder and faster you pedal, the greater the force and the faster the bicycle goes.
- Change direction – a force can cause a moving object to change direction e.g. a tennis ball approaching a tennis player, the player hits the ball and it changes direction because the racket exerts a force on the ball to change direction
- Change in position – a force can cause a stationary object to move and change position e.g. if a soccer ball is kicked, the foot exerts the force on the ball and the ball moves forward.

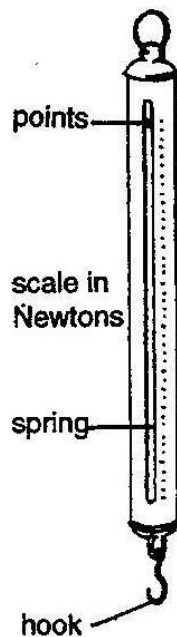
Types of force

- There are two main types of forces i.e. contact and non-contact forces
- Contact forces are forces that are in direct contact with each other e.g. pulling, pushing and twisting forces (mechanical forces) that are directly applied to an object.
 - Weight – is the force acting on mass due to gravity
 - Mechanical force – caused by the movement of objects e.g. falling water
 - Friction – force that opposes motion. It occurs between two objects that are moving against each other. It slows down a moving object or causes it to stop.

- Non contact forces that not in direct contact with each other. Effect of the force can be observed e.g.
 - Gravitational force – that pulls objects towards the centre of the earth.
 - Magnetic force – force exerted by magnets to attract or repel materials
 - Electrostatic force – produced by rubbing and can attract or repel objects

MEASURING FORCE

- Is measured in Newtons (N) using a force meter or spring balance
- A forcemeter is made of a spring with a hook attached to it. The larger the force applied, the longer the spring stretches and the greater the reading. The scale of a forcemeter is in Newtons.
- A spring balance is a type of forcemeter that measures how much a spring stretches when an object is hung from it. This is the object's mass. The mass reading on a spring balance can be converted to weight by multiplying the mass by the force of gravity (about 10N per kg) weight is measured in Newtons



$$100\text{g} = 1\text{N}$$

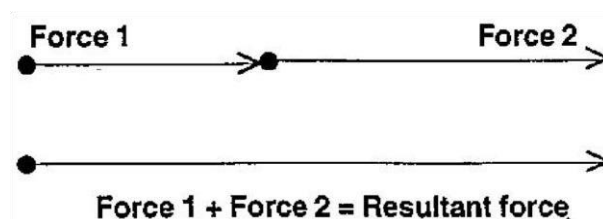
$$1\text{kg} = 10\text{N}$$

BALANCED AND UNBALANCED FORCES

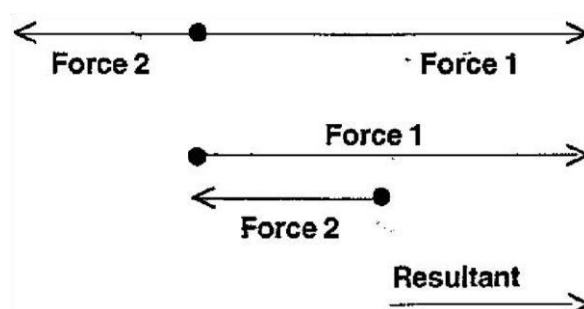
- If two equal sized forces acting in opposite direction are balanced, no movement will occur thus the forces are in equilibrium.
- If two forces are unbalanced then movement will occur in the direction of the larger force.

Resultant force

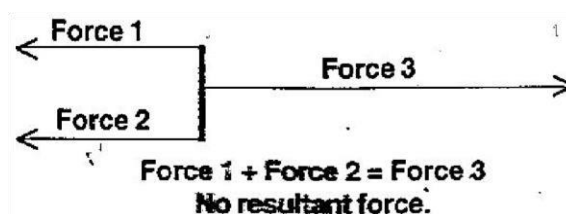
- Is the sum or difference of all forces acting along the same line
- When forces act along the same line and in the same direction their resultant is the sum of the individual forces



- When forces acting in the same line but in opposite direction their resultant is the difference of the individual forces.



- The resultant force on a body causes movement in its direction. If the resultant is zero, the body is at equilibrium and does not move i.e. equal force in opposite directions.



- Direction in which the force is acting is shown by arrows and if drawn to scale they indicate the size of the force.
- The resultant force on a body cause movement in its direction

Example 1 A small cart is pulled by 2 oxen pulling with 200N. The force of friction between the cart and the ground is 50N and between the wheel and axle is 75N. Find the resultant force.

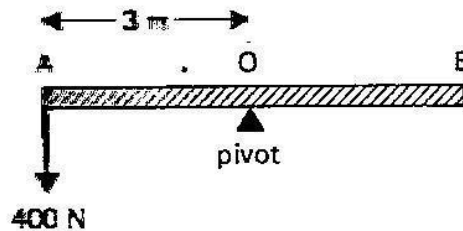
Example 2 A wheelbarrow is pushed along a force of 150N. The force of friction between wheel and the ground is 30N. Draw a diagram to show the forces acting and

hence find the resultant force.

MOMENT OF A FORCE

- Moment is the turning effect of a force about a fulcrum (pivot)

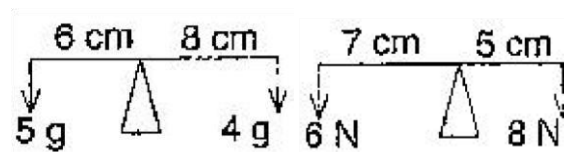
- It depends on both the size of the force and how far it is applied from the pivot
- It is measured by multiplying the force by the perpendicular distance of the line of action of the force to the fulcrum.
- In the diagram below, the moment, M , of the force P is given by $M = Fx$



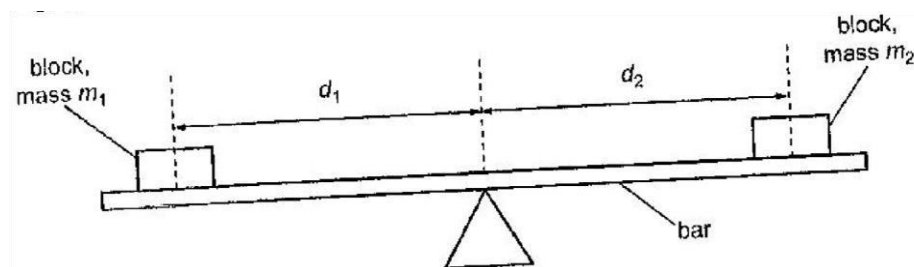
Moment of a force = force applied $\times$ distance from turning point

- Moment is measured in Newton metres (Nm)
- The greater the force and the distance from the fulcrum, then the greater the turning effect.

Examples Calculate the clockwise and the anticlockwise moments for each of the following examples below.



The principle of Moments

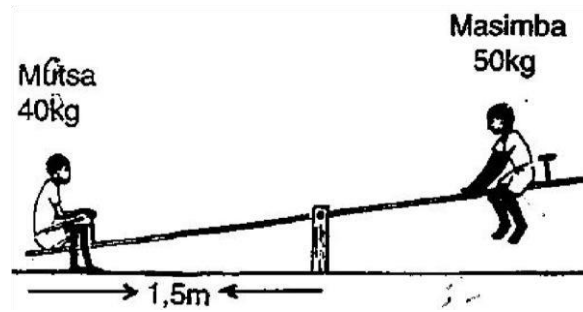


- The principle is about objects balancing when the moments of all the forces acting are balanced.
- The principle states that for a body to be in equilibrium (balanced), the sum of the moments about any point is zero i.e. clockwise moment = the anticlockwise moments

$$F \times d \text{ (anticlockwise)} = F \times d \text{ (clockwise)}$$

Example 1 Mutsa who has a mass of 40kg, sits at one end of a see-saw that is 3m long.

The see-saw pivots at its centre point. Where must Masimba, who has a mass of 50kg, sit to balance Mutsa.



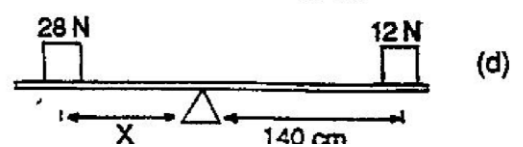
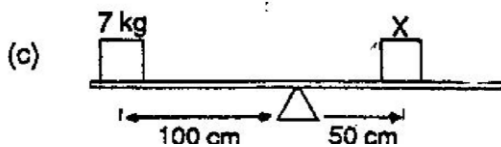
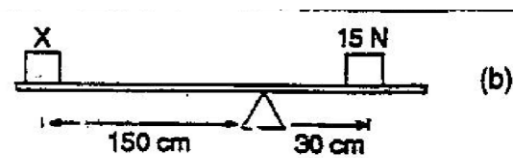
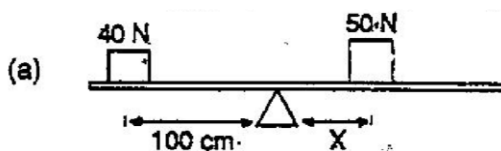
By the law of moments,

Clockwise moments = anticlockwise moments

$$\therefore 40\text{kg} \times 1.5\text{m} = 50\text{kg} \times X\text{m}$$

$$\therefore X = \frac{40 \times 1.5}{50} = 1.2\text{m}$$

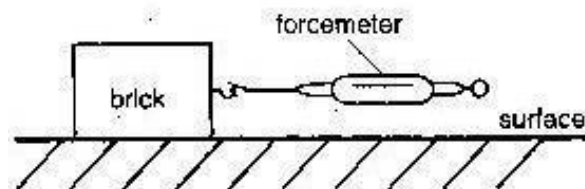
Example 2 Use the principle of moments to calculate the unknown quantities in examples below. In all cases, the beam is balanced.



FRICTION

- Friction is a force that opposes motion (movement) when two surfaces are in contact with each other. Friction occurs between two objects that are moving against each other
- An object remains stationary if frictional force is greater than pushing force - An object only moves if pushing or pulling force is greater than friction.

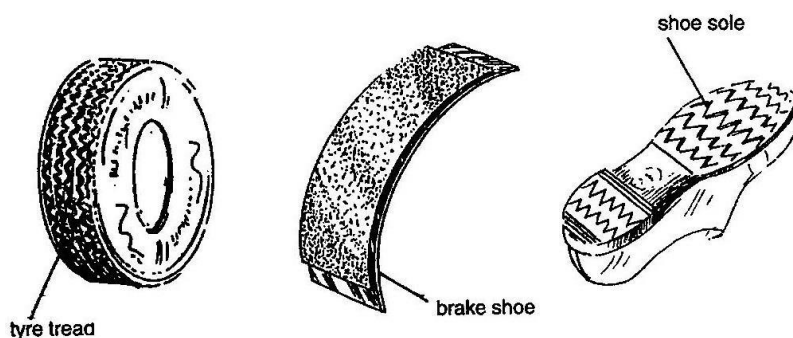
Measuring friction



- Tie a string around a brick and hook the force meter to the string. Use the string to pull the brick.

- Start with a small force and continue to increase the force, noting the reading on the scale of the force meter as you increase the pulling force. Record the maximum force reached before the brick starts moving.
- Continue to pull the brick so that it slides with minimum force. Record the reading on the force meter while the block is moving.
- As the brick is pulled forward, it didn't start moving as soon as it is started pulling. Another force acting backwards occurs at the same time i.e. friction which exists between two surfaces that are in contact.
- This opposing force has a maximum size for any two surfaces in contact such that when this size is exceeded, the object starts moving.
- The maximum force recorded by the forcemeter just before the brick starts moving is called static friction.
- As soon as the brick starts moving a smaller force is required to keep brick moving.

Application of frictional forces



Nature of surface – the rougher the surface, the greater the friction it will cause.

Road surfaces – are rough to improve friction

Car breaking systems – the force of the foot on the brake pedal is multiplied by the breaking system. There is a large force at the brake pads against the wheel and the force of friction stops the car.

Tyre threads – the threads on tyres provide a rough surface to improve friction between the wheel and the road surface. This improve grip on the road and prevents skidding.

Shoe soles – are rough to increase friction and give a better grip

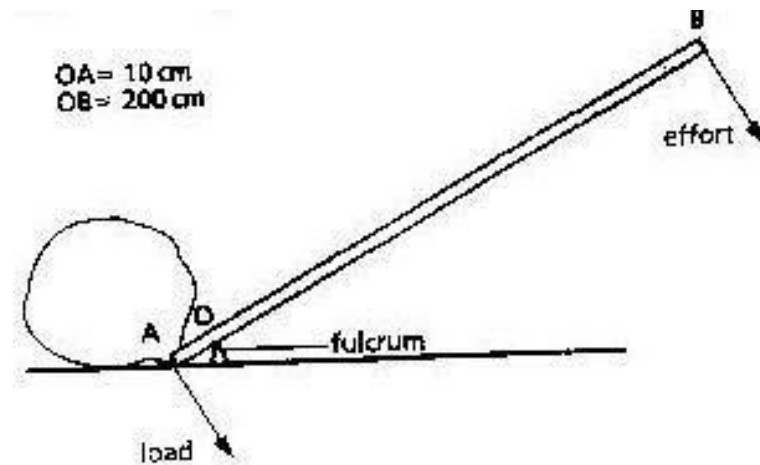
- All these increase the roughness of surfaces to control motion and remain safe - Friction can be reduced by
 - Lubricating
 - Use of rollers
 - Use of ball bearings

- Polishing surfaces
- Adding wheels

SIMPLE MACHINES

- A machine is a device that helps us to do work easier e.g. levers, pulleys, inclined planes, gears, wheel and axle.
- A machine is anything that magnifies a small force into a large force that is able to do work.
- Machines are energy convertors

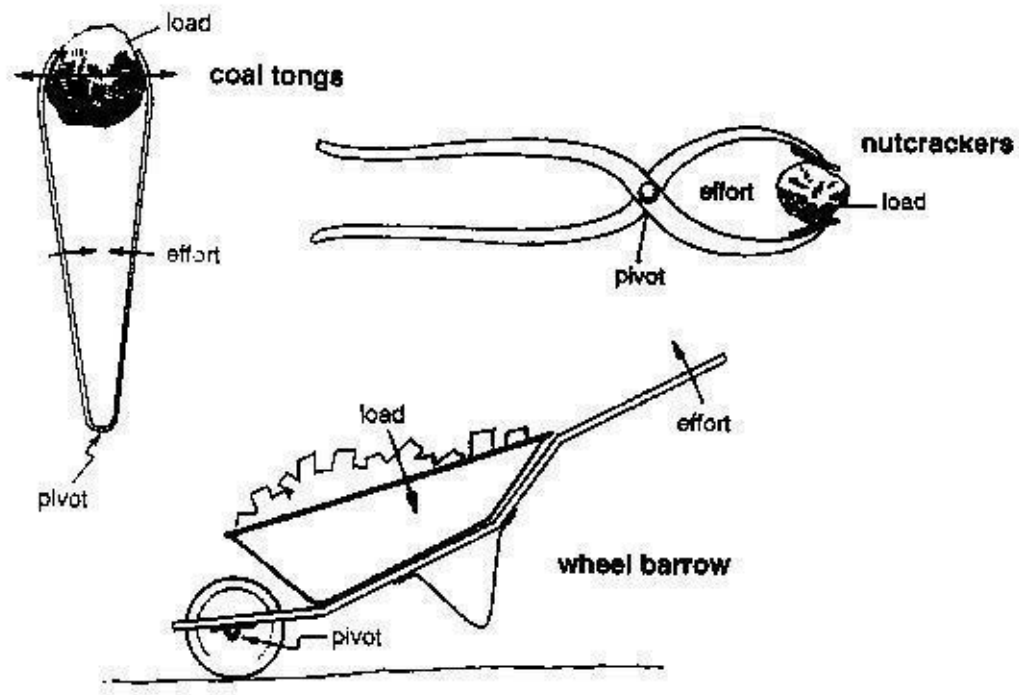
Levers



- A lever is a device which can turn about a pivot
- In a working lever a force called the effort is used to overcome a resisting force called the load. The pivotal point is called the fulcrum
- If we use a crow bar to move a heavy boulder, our hands apply the effort at one end of the bar and the load is the force exerted by the boulder on the other end.
- A lever changes the force applied on one end of the lever to an opposite force on the other side

Classes of lever

- First class: fulcrum is between the load and effort e.g. crowbars, scissors, claw hammer, pliers.
- Second class: load is between the fulcrum and effort e.g. wheelbarrow, nutcracker, bottle opener
- Third class levers: effort is between the fulcrum and load e.g. a hoe, fishing rod, tongs, a spade



ENERGY

EFFECTS OF ENERGY

- Energy is the ability to do work
- Effects of energy are things that can be seen as a result of energy e.g.
 - Heating magnesium – burns brightly i.e. gives off much light
 - Squashing a spring – spring changes shape i.e. becomes shorter
 - Pulling elastic band and pluck it – elastic vibrates and make sound
 - Bouncing a ball – ball changes direction i.e. up and down
 - Blowing a whistle – sound comes out
 - Closing switch on an electric motor – motor turns
 - Clapping hands, sound is produced
 - Lighting a torch, chemical energy is converted to light energy
 - Raising a brick, chemical energy is converted to kinetic energy and potential energy
 - Car engines release stored energy and convert it into motion

Sources of Energy

- A source of energy can supply the energy that can be used to do work. - Source of energy include
 1. Renewable sources i.e. wind, bio-fuels, solar, hydropower
 2. Non-renewable i.e. fossil fuels, nuclear energy

FORMS OF ENERGY

Kinetic energy

- Energy possessed by a body by virtue of its motion e.g. when kicking or bouncing a ball, energy is used to move the ball. The ball has kinetic energy. All moving objects have kinetic energy. The faster it moves the more kinetic energy it has.

Heat energy

- Heat energy is also called thermal energy and is the final fate of other forms of energy. Heat energy causes a change in the internal energy. It is transferred by conduction, convection or radiation. The greater the heat energy of a substance is, the higher its temperature.

Electrical energy

- Energy formed when electrons flow through a conductor. It can be changed into many other forms like heat and light energy. It is produced by energy transfers at power stations and in batteries

Chemical energy

- Chemical energy is energy released if a chemical reaction takes place. The energy of food is released by chemical reactions in our bodies during respiration. Fuels cause energy release when they are burnt. Batteries are compact sources of chemical energy, which in use is converted to electrical energy.

Potential energy

- Potential energy is the energy a body has because of its position or condition e.g. water stored in a reservoir has potential energy stored in the form of gravitational potential energy. It is therefore stored up energy that can be released to do work - There are different types of potential energy i.e.

- *Gravitational potential energy* – is the energy stored in a body due to its relative position to the earth. For example, water stored behind a dam wall, it has more potential energy than water at lower level. When released, the water loses potential and gains kinetic which can be used to drive machinery which generate electricity
- *Elastic potential energy* – is the energy stored by a stretched or squeezed body. For example, a stretched bow or catapult has elastic potential energy which can be used kinetic energy to a missile. Similarly, a wound up spring can drive a watch. The more an object is stretched or squeezed, the greater its elastic potential energy.
- *Chemical potential energy* – is the energy within objects that will be converted if a chemical reaction takes place. Food and fuels are stores of chemical energy. Foods release

energy as a result of chemical reactions in the body. When fuels are burnt they release energy mainly as heat. The heat is used to drive an engine.

Light energy

- Light energy is the energy that enables us to see and is important for green plants to make food by photosynthesis. Main source is the sun.

Sources of light energy

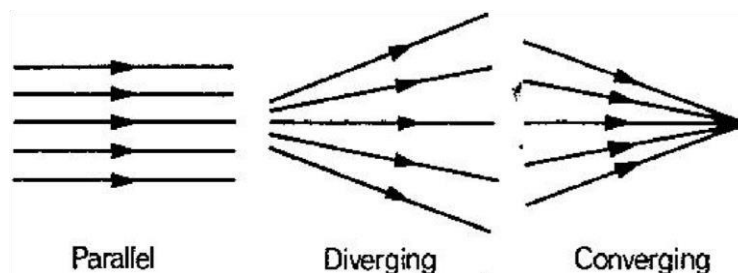
- Sun
- Moon
- Fire
- Electric bulb
- Candle
- LEDs

Transmission of Light

- A ray of light is a narrow stream of light and is shown by straight lines with arrows showing the direction in which the light is travelling.



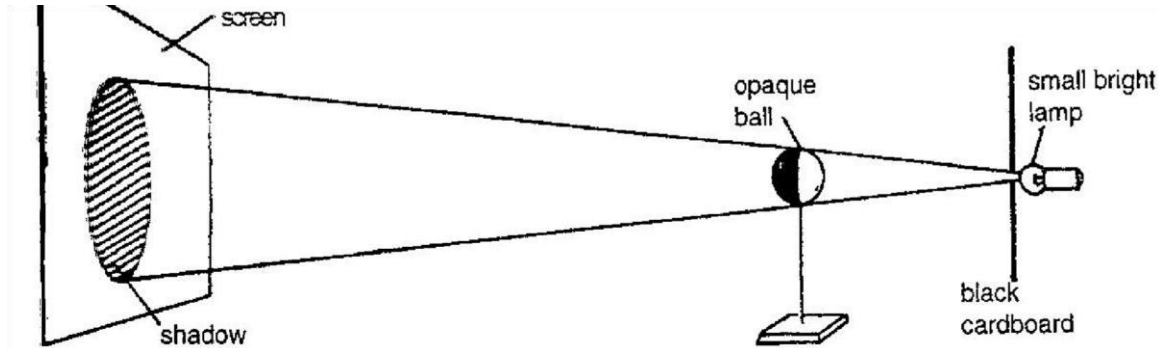
- A beam of light is a stream of light usually represented by a number of rays. A beam can be parallel, divergent or convergent as shown below



Production of shadows

- A shadow is an area in which light does not fall
- Shadows are formed because some (opaque) objects do not allow light to pass through them and that light travel in a straight lines. Sharpness of the shadow depends on the size of the light source.

- If light is made to pass through a narrow hole, a straight ray of light is produced. This ray has defined edges. This is because light travels in a straight.
- If the light is blocked by a figure with sharp edges, a sharp shadow of the object with well defined edges is cast onto a screen.



Sound energy

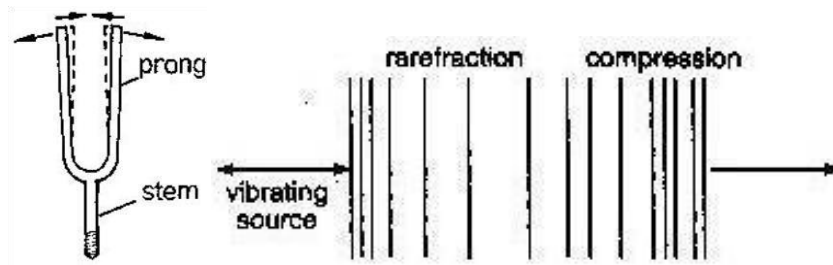
- Sound energy is energy produced when an object or substance vibrates and it moves through materials as sound waves.

Production of sound energy

- Sound is produced by vibrations.
- Vibrations are small but fast backward and forward movements.
- If an elastic band is plucked, it will be seen to vibrate and it will produce a humming sound. The elastic band represents the tissue called the vocal chords found in the throat which vibrate to produce sound.
- Musical instruments make sound waves when part of them or the air inside them vibrates

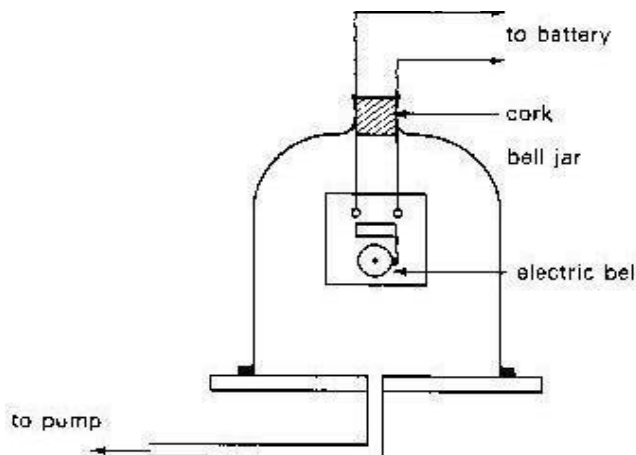
Transmission of sound energy

- Sound waves are mechanical in nature in that they require a medium through which to move which is usually air, liquids or solids.
- Sound are transmitted when particles of a medium vibrate back and forth in the same direction
- When these molecules vibrate, they in turn cause the molecules around them to vibrate causing the sound to be transmitted
- Diagram shows how a vibrating tuning-fork sends out a sound wave. When the tuning fork moves to the right it compresses the air particles together. This disturbance is then transmitted from particle to particle through the air, with the result that a pulse of compression moves outwards.



- A series of compressions and rarefaction moves sound forwards, but the particles of the medium only move backwards and forwards. This is why sound needs a medium in order to be transmitted.

Experiment to show that sound requires a material medium for transmission



- The bell-jar experiment can be used to demonstrate that sound waves cannot pass through a vacuum.
- Before starting the pump, switch on the electric bell. You can see the movement of the striker and hear the sound at the same time.
- When the vacuum pump starts, the sound becomes fainter and fainter with time. Finally it becomes so soft that you cannot hear it any more although the striker can still be seen hitting the gong. If you let air into the bell jar, sound is heard again. This shows that without a medium sound cannot be transmitted.
- Care must be taken that the bell should not touch the glass and the connecting wires used must be thin to prevent any sound waves from being transmitted through them to the outside of the jar as the striker vibrates vigorously

ENERGY CONVERSIONS

Conservation of energy

- The law of conservation of energy states that the total energy of a closed system is constant i.e. energy is neither lost nor gained but simply transformed from one form to another.

- The total energy of a closed system will be same before an interaction as after it
- When energy is transformed from one form to another either work is done or energy is used.

Energy convertors

- Torch
- Dynamo
- Catapult
- Solar panel
- Bulb
- Green plants

Energy chains

- **Energy chain** is a diagram that shows energy conversion.
- Energy can be converted from one form to another

Green plants solar energy from the sun → chemical energy in carbohydrates

Catapult chemical energy → kinetic energy → potential energy → kinetic energy

Dynamo kinetic energy of dynamo → electrical energy → light energy

Bulb electrical energy → heat and light energy

Torch chemical energy from cells → electrical energy → light energy

Solar panel solar energy → electrical → chemical energy in cells

WORK AND ENERGY

Energy

- Is the capacity to do work -

SI units is the Joule (J)

- The amount of work done should be equal to the amount of energy used.

Work

- Work is done when a force moves. It is a measure of the amount of energy transferred
- The amount of work done depends on the size of the force applied distance and distance it moves

- Work is a measure of energy expended in moving an object.

Energy used = Work done

work done = force $\times$ distance moved in the direction of force

- The unit of work is the joule (J) or Newton metre (Nm)
- One joule it is the work done when a force of one Newton (N) moves through one metre
- For example; if you have to pull with a force of 50N to move a crate steadily 3m in the direction of the force, the work done is

$$50N \times 3m = 150N = 150J$$

Example 1 How much work is done when a weight lifter lifts eight 100N boxes through a height of 0.50m?

Example 2 A girl lifts a pile of books with a total mass of 5kg through a vertical distance of 0.75m. How much work does she do?

MAGNETISM

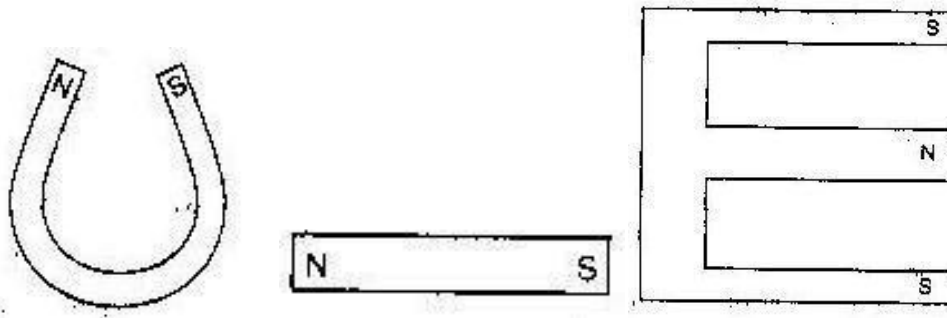
MAGNETS

- Magnetism refers to magnetic force caused by the unique properties of certain materials.
- Are usually made of metal iron or steel

Properties of magnets

1. Can pick magnetic materials
2. Has two poles i.e. north and south pole
3. It has a magnetic field
4. Exerts repulsive and attractive force

Types of magnets



Bar magnet – consist of a straight bar of magnetic material with one end being north and the other south pole. This shape may result in a weak magnetic force as the magnetic force is weak on the sides and is concentrated at the ends of the bar magnet. Can be used in refrigerator doors

Horse shoe magnet – are bar magnets that bent in a U-shape. The N and S poles point in the same direction resulting in a strong magnetic force that is stronger around both poles. Magnetic force of both poles is felt. It can be used for lifting any metal objects depending in size and strength.

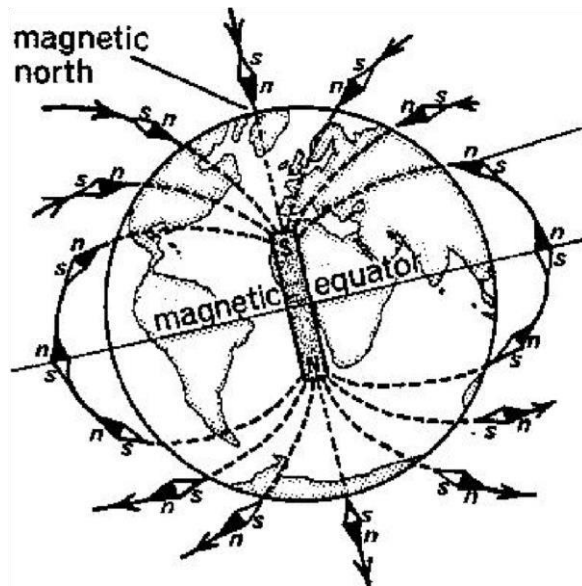
C magnets – are shaped like C, in an arc. They are used to create a magnetic force in a motor.

Used in washing machines, fridges, speakers, air conditioners, generators etc

E (Electromagnets) – magnetic field is caused by an electric current. It consists of a coil wire wound around a soft iron core. The wire is connected to a power source. When current flows through, it causes a magnetic field that magnetises the iron core

The earth as a magnet

- If lines of force are plotted on a sheet of paper with no magnets near, a set of parallel straight lines are obtained. They run roughly from S to N geographically and represent a small part of the earth's magnetic field in a horizontal plane.

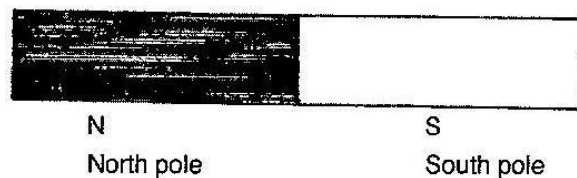


- The earth has opposite poles i.e. the north-pole at the top of the globe and south-pole at the bottom of the globe. It is surrounded by a magnetic field.
- The direction of the earth's magnetic field lines is from the magnetic north pole to the magnetic South Pole.
- The earth's magnetic pole and geographical poles are opposite. The earth's magnetic south pole is where the geographical north pole is and the earth's magnetic south pole is where geographical north pole is

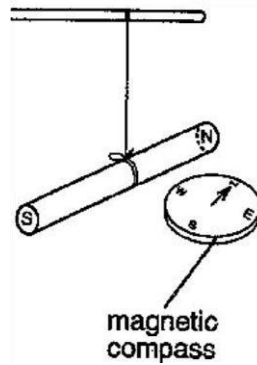
Magnetic from non magnetic materials

- Magnets can attract some objects. All materials attracted by magnets are called magnetic materials e.g. iron, nickel, cobalt
- All materials that are not attracted by a magnet are called non magnetic materials e.g. wood, rubber, plastic, glass, copper, aluminium.

Magnetic poles

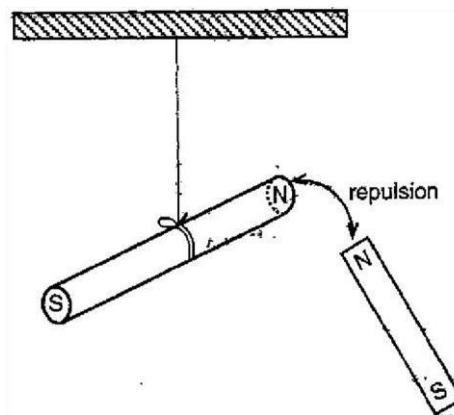


- Magnets have two poles i.e. north and south poles. Magnetism is concentrated around the poles (ends) of a magnet.
- Poles of a freely suspended magnet that always rest pointing towards the north is the north pole(N) while the one that points to the south is the south pole (S)



- Magnetic force acts mainly at the poles of a magnet - To determine the polarity of bar magnets;

- Take one bar magnet and hang it freely on a retort stand
- Bring one pole of the second magnet close to the hanging magnet. Observe what happens.
- Turn the second magnet so that the poles face the opposite and hold it close to the hanging magnet. Observe what happens.



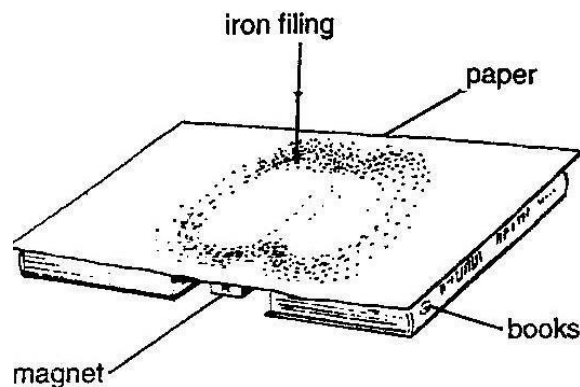
- If the N pole of a magnet is brought near the N pole of another magnet, repulsion occurs. Two S (south seeking) poles also repel. By contrast, N and S poles always attract.
- The law of magnetic poles summarises these facts and states:

Like poles repel, unlike poles attract.

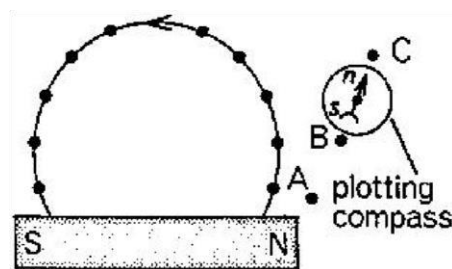
- The force between magnetic poles decreases as their separation increases.
- To test whether an object is a magnet, bring one end of this object towards one end of a suspended bar magnet. If repulsion occurs, the object is a magnet since repulsion occurs between like poles
- A given sample is magnetic only if one of its ends repels a magnet. Attraction is not used to verify a magnet because any magnetic material is attracted even if it is not a magnet.

Magnetic fields

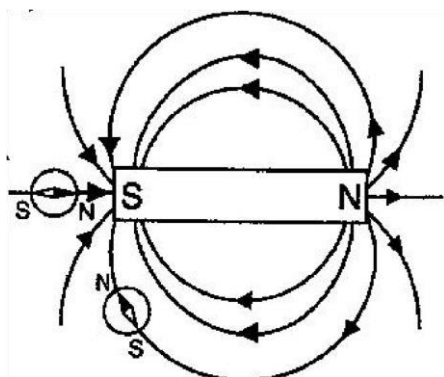
- A magnetic field is the region where magnetic force is exerted on any magnetic object placed within the influence of the field.
- Magnetic field lines show the direction and strength of the magnetic field around a magnet. The direction of magnetic field is always from North Pole to South Pole.
- Magnetic force is strongest at the poles of a bar magnet where field lines are most concentrated that is at the poles.
- To show the pattern of a magnetic field around a bar magnet, place a sheet of paper on top of a bar magnet and sprinkle iron filings lightly and evenly on to the paper. Tap the paper gently with a pencil and the filings should form patterns showing the lines of force. The iron filings will line up along the magnetic field lines.



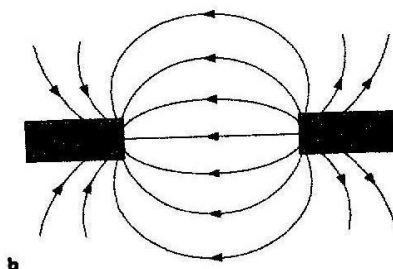
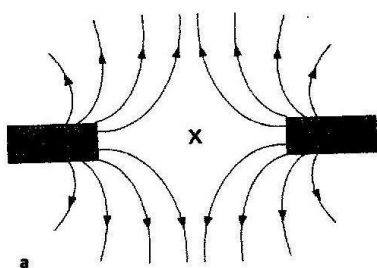
- The plotting compass can also be used to plot the combined magnetic field due to neighbouring magnets or around a bar magnet.
- Lay a bar magnet on a sheet of paper. Place the plotting compass at a point such as A, near one pole of the magnet (north pole)
- Mark the position of the poles (n; s) of the compass by pencil dots. B, A
- Move the compass so that the pole s is exactly over B, mark the new position of n by dot C
- Continue this process until the other pole of the bar magnet is reached. Join the dots to give one line of force and show its direction by putting an arrow on it. By convention, the field direction draws north to south.



- Plot other lines by starting at different points around a bar magnet as shown below



- Magnetic field gets weaker as the distance from the magnet increases – the lines are further apart
- The field lines also help explain what happens as two magnets are brought together – the field lines will interact with each other giving either repulsion or attraction, depending on the direction of the interacting field lines

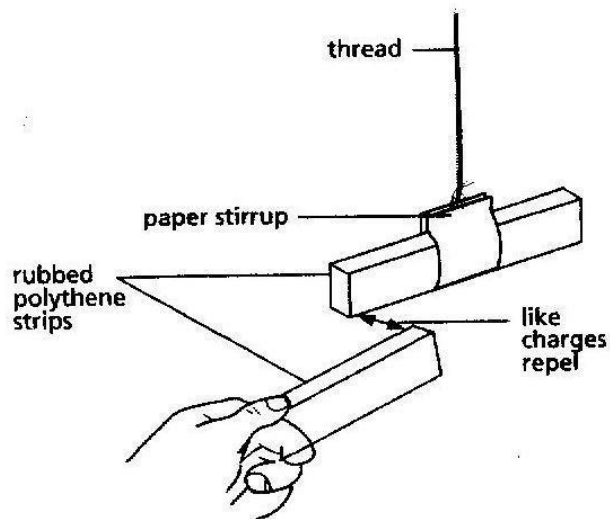


ELECTRICITY

STATIC ELECTRICITY

Charges

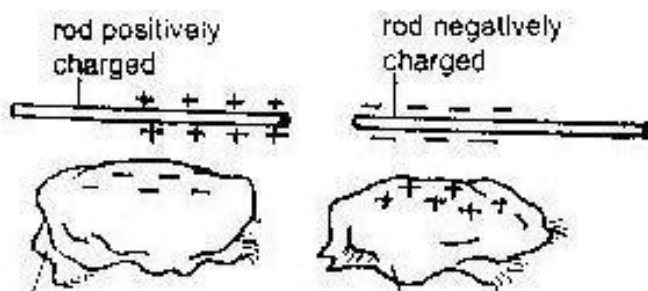
- When a strip of polythene is rubbed with a cloth it becomes charged. If it is hung up and another rubbed polythene strip is brought near, repulsion occurs. Attraction occurs when a rubbed strip of Perspex is brought near.



- This shows that there are two kinds of electric charges i.e.
 - **positive (+)** (on the perspex rod)
 - **negative (-)** (on the polythene rod)
- It shows that like charges repel, while unlike charges attract.

The production of charges

- Atoms are made up of positively charged protons, negatively charged electrons and neutrons that have no charge.
- If an object gains electrons, it will become more negatively charged and if it loses electrons, it will become more positively charged.
- Rubbing a rod using a woollen cloth produces an electrical charge. The production of charges by rubbing can be explained by supposing that electrons are transferred from one material to the other. Either the electrons are removed from the rod or electrons are added onto the rod.



- Rubbing polythene with a cloth makes it become negatively charged i.e. electrons are added onto the rod
- Rubbing perspex with a cloth makes it become positively charged i.e. electrons are removed from the rod.

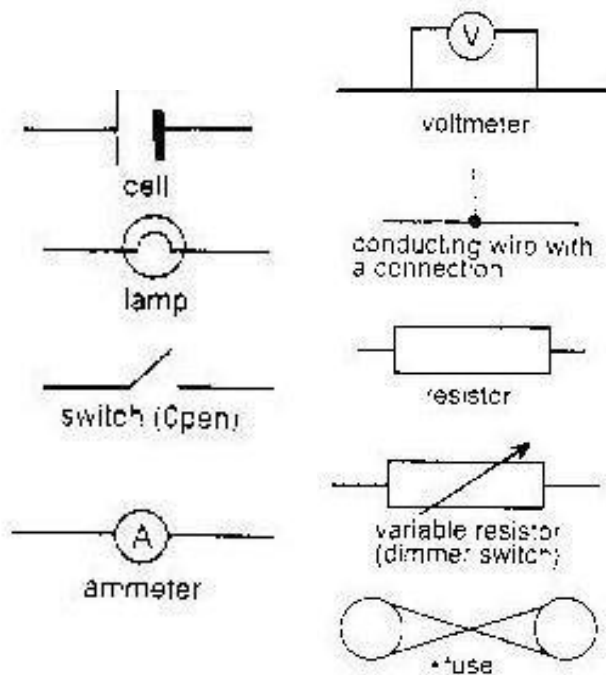
Note that it's only electrons which move; the protons remain fixed in the nucleus.

CURRENT ELECTRICITY

Simple d.c. circuits

- It is important to draw simple and clear diagrams. The following diagrams give a list of special symbols that are used to represent common devices that are usually employed in electric circuits

Circuit symbols

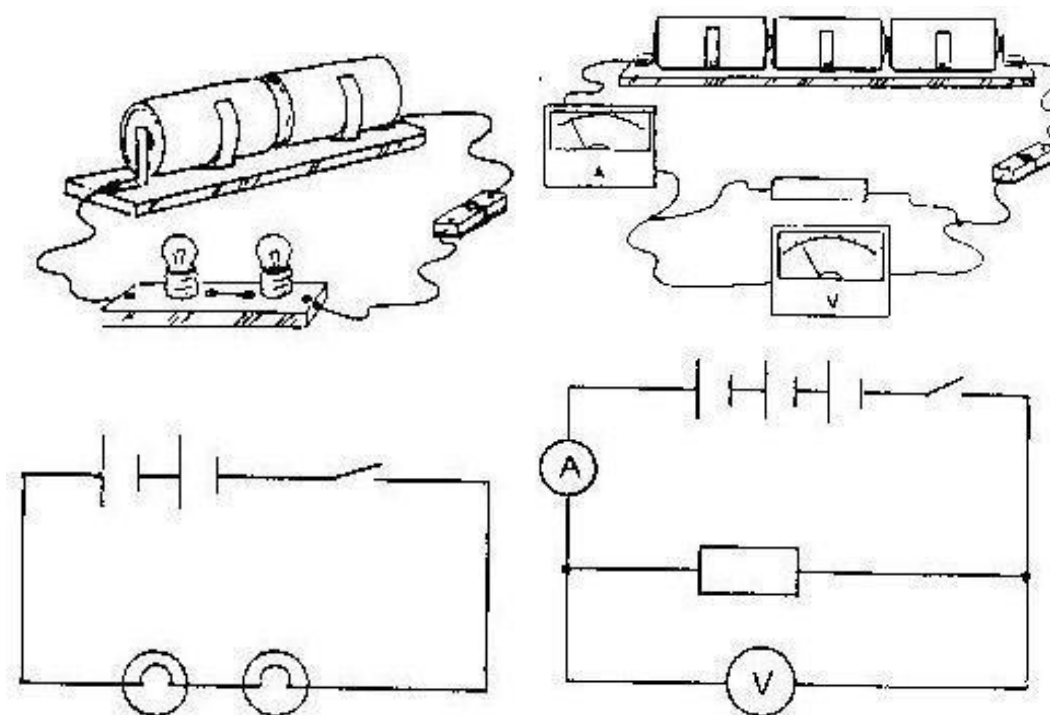


Components of direct current circuit

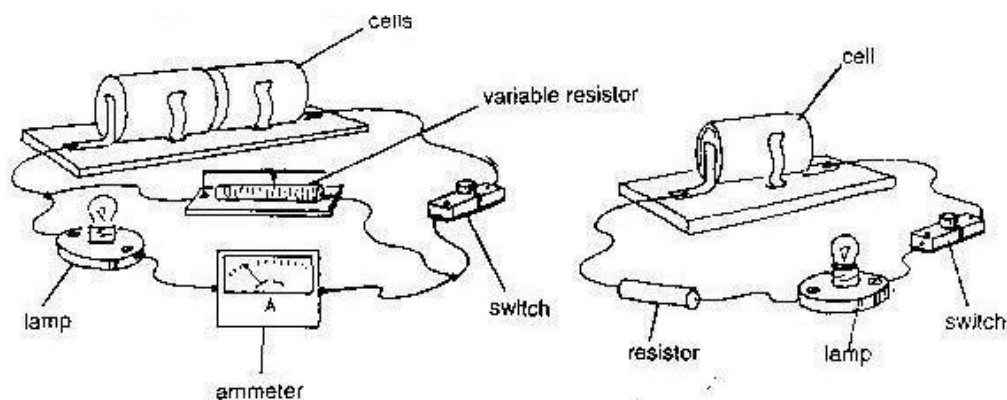
- The cell is the source of electrical energy. The longer line is the positive terminal and shorter and thicker line is the negative terminal of the cell.
- The switch breaks or completes an electric circuit, stopping or starting the current flow. When switch is open, circuit is incomplete and current does not flow. When the switch is closed the current flows since the circuit is complete.
- A resistor is a device that offers opposition to the flow of current. It restricts the flow of current in circuits, slowing the current down.
- A light bulb is a resistor that changes electrical energy into light and heat energy.
- A fuse is a device that protects appliances and users from electrical shocks
- An ammeter is used to measure the amount of current
- A voltmeter is used to measure voltage of the circuit

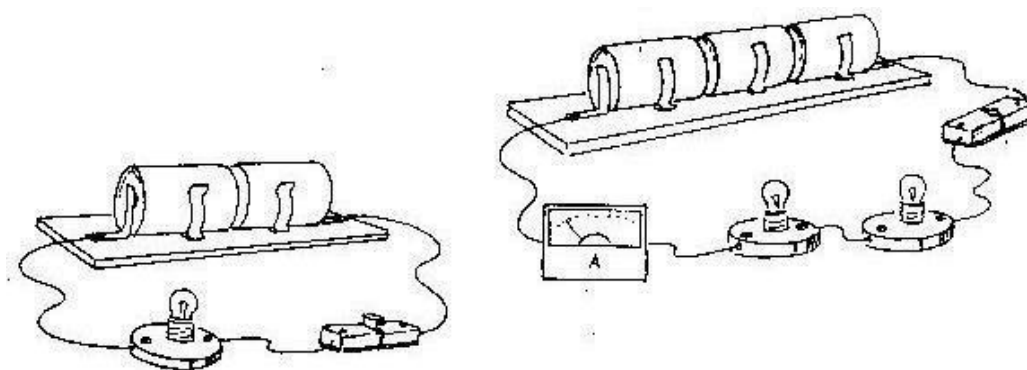
- A variable resistor regulates or controls the current in a circuit i.e. it can increase or decrease the amount of current.
- Connecting wires provide path for electric current
- *Every circuit has a power source, switch and a load (all components that use the electrical energy and convert it to other forms)*

Representing some circuit diagrams using electric symbols



Draw a standard circuit diagram using symbols for each of the circuits shown below

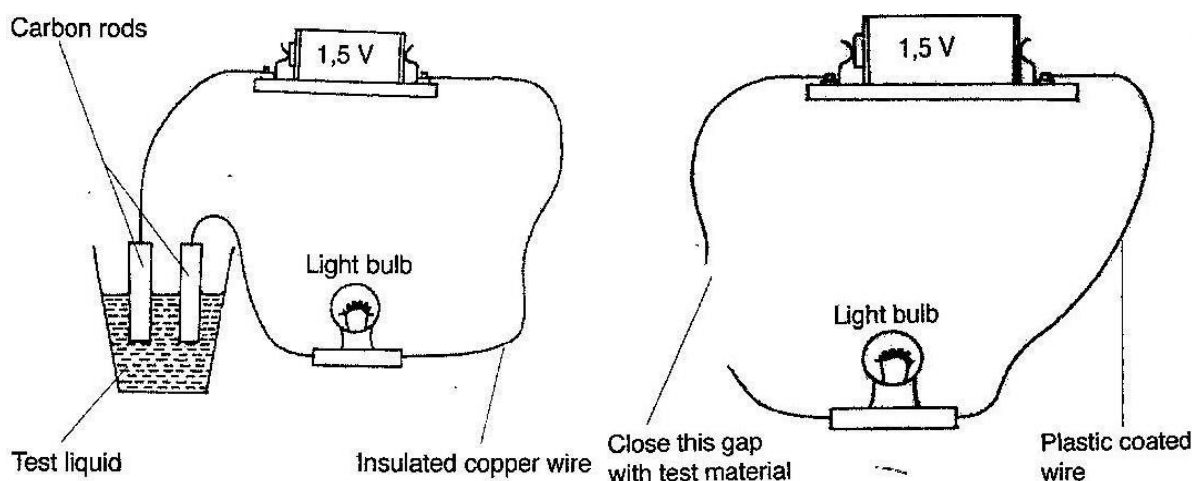




CONDUCTORS AND INSULATORS

- Conductors are materials that allow electricity to flow through it e.g. copper, carbon, and salt water.
- Insulators are materials that do not conduct an electric current e.g. rubber, plastic, wood, glass, pure water - A solution that conducts electricity is called an electrolyte. The solution is chemically changed by the current
- Conductors are generally used for transmission of electricity.
- Insulators generally protect people from electrocution
- Most metals are conductors while non metals are insulators except for graphite (carbon).
- Gases are also very poor conductors of electricity

Experiment on conductivity of different materials



- Make a simple circuit using a battery connected to two wires and a light bulb as shown in the diagram.

- Connect different materials using crocodile clips to find out which ones conduct or do not conduct electricity. If they do conduct electricity the light bulb will light up.
- Record results in the table below

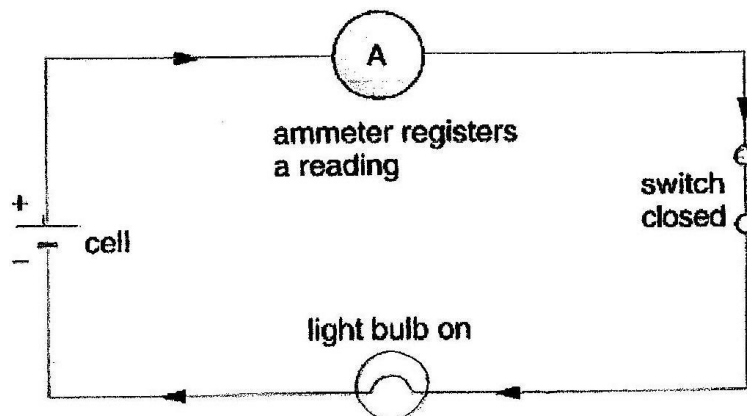
Material	Observation	Conclusion

CURRENT, VOLTAGE AND POWER

Current

- Is the flow of charge in an electric circuit
- An ammeter is used to measure current
- The unit of current is the ampere (A) or amps

Measuring current



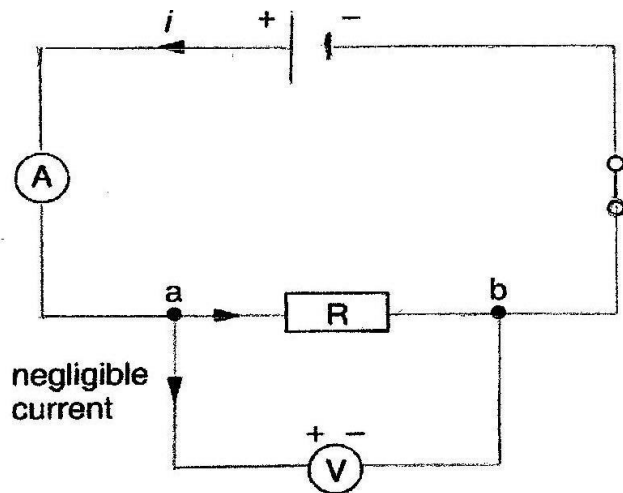
- An ammeter is always placed in series with the resistance or other circuit components through which the current is to be measured.
- Ammeters should therefore have low resistance compared with that of the rest of the circuit, so that they do not introduce unwanted resistance

Voltage

- Is the measure of the ability of electrical energy to do work.
- It is measured in volts (v) using a voltmeter
- It is supplied by the cell or battery

Measuring voltage

- A voltmeter is always placed in parallel with the resistance or apparatus across which the potential difference has to be measured.
- Voltmeter ought therefore to have a high resistance compared with the resistance across which the voltage is to be measured, so that they take a comparatively negligible current, and so disturb the circuit as little as possible.



Power

- Electrical power *is* the rate at which electrical energy is changed to other forms of energy in an electric circuit.
- Power is measured in watts (W).

$$\text{power} = \text{voltage} \times \text{current}$$

$$P = VI$$

Example 1 What is the power of an electric light bulb if it is drawing a current of 0.25A from the mains electricity supply of 240V?

Example 2 What current is drawn by a 1.5kW heater which operates on a 240V mains supply?

Example 3 What voltage is needed for a 0.5A current to pass through a 100W light bulb

